Pricing and Hedging Deribit Inverse Options under Affine Stochastic Volatility Models with Jumps

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Abstract

This thesis studies the pricing and hedging of Deribit inverse options under affine stochastic volatility models with jumps. Unlike standard cash-settled options, inverse cryptocurrency options have premia and payoffs denominated in the underlying coin while strikes are quoted in USD, so valuation and hedging must account explicitly for contract denomination and numeraire choice. To do so, the thesis combines the inverse-option framework based on pricing under the futures numeraire and premium-adjusted net delta with three pricing models: Black, Heston, and a stochastic volatility model with correlated jumps in returns and volatility (SVCJ). The models are calibrated to BTC and ETH Deribit option panels using characteristic-function methods and the Carr–Madan FFT procedure, and are evaluated using both out-of-sample pricing and interval-level hedging tests with inverse perpetual futures.

The results show that SVCJ delivers the best out-of-sample pricing performance, particularly in harder regions of the surface such as short maturities and the extreme moneyness. Hedging results are more mixed: dynamic hedging reduces risk substantially relative to an unhedged position, but the main improvement comes from using the premium-adjusted net delta rather than the conventional delta, while differences across Black, Heston, and SVCJ are small over short rebalancing intervals. The thesis therefore concludes that richer stochastic-volatility-with-jumps dynamics improve pricing, whereas short-horizon delta-hedging performance depends primarily on correct inverse-option hedge design and remains constrained by basis risk between the underlying dated futures and the perpetual futures used for hedging.

Contents

Introduction

Cryptocurrency derivatives markets have developed around contract forms that differ in economically important ways from the conventions familiar from equity options. Among these instruments, inverse options occupy a central place, especially on Deribit, where option premia and payoffs are denominated in the underlying coin while strikes are quoted in USD. This contract design is not a superficial change of units. Once the premium and payoff are settled in coin, the valuation problem, the interpretation of Greeks, and the practical construction of a hedge all differ from the standard setting of a cash-settled European option. For this reason, pricing and hedging inverse options should not be treated as a routine extension of standard Black–Scholes type machinery. They require a framework that is simultaneously consistent with exchange conventions, mathematically coherent, and sufficiently flexible to capture the stylized facts of crypto option markets.

These stylized facts make the problem especially demanding. Cryptocurrency returns are well known to exhibit high volatility, heavy tails, abrupt jumps, and substantial time-variation in volatility. In option markets, these features appear as pronounced smiles and skews, short-end distortions, and cross-sectional patterns that are difficult to reconcile with a constant-volatility diffusion. A model that ignores these features may still serve as a useful benchmark, but it is unlikely to provide a satisfactory description of observed option surfaces across strikes and maturities. The central topic of this thesis is therefore the pricing and hedging of Deribit inverse options under affine stochastic volatility models with jumps, with particular emphasis on whether a richer crypto-specific specification improves upon standard benchmark models in both valuation and risk management.

The existing literature provides strong building blocks for such a study, but it does not yet fully combine them in the form required by the market under consideration. On one side stands the inverse-option literature. Lucic and Sepp (2024) develop the valuation logic that links regular USD-settled options to inverse coin-settled options and show that the relevant hedge ratio for inverse contracts is not the conventional delta, but a premium-adjusted net delta. Their framework is essential because it clarifies that inverse options should be analysed under a futures-based numeraire and that the option premium itself enters the delta hedge. Alexander et al. (2023) complement this perspective by interpreting inverse crypto options through an FX- and quanto-style lens, highlighting the importance of denomination, collateralisation, and market incompleteness. This literature explains how inverse contracts should be valued and hedged in principle, but it does not provide a rich framework for the valuation of Deribit inverse options that accounts for the stylized facts of cryptocurrency returns and volatility mentioned above.

On the other side stands the literature on cryptocurrency option pricing under richer return dynamics. High-frequency and option-implied evidence points to the importance of jumps in returns and volatility, while empirical studies of the Bitcoin and broader crypto markets show that purely diffusive models struggle to reproduce the cross-sectional structure of implied volatility. In that literature, the stochastic volatility model with correlated jumps in returns and volatility proposed by Hou et al. (2020) is especially relevant for the present thesis. The model belongs to the affine class, retains tractability and numerical efficiency through characteristic-function pricing methods, and introduces exactly the sort of co-jump structure that is likely to matter in crypto markets, particularly for short maturities and far-from-the-money strikes. However, that literature is focused on direct options in settings that do not explicitly match the inverse payoff structure and hedging conventions of listed Deribit options on futures. The resulting gap is clear: the inverse-option literature gives the right valuation identity and hedge concept, while the crypto stochastic-volatility literature gives the right family of underlying dynamics, but the two strands are not fully integrated in an empirical study of Deribit inverse options.

This thesis addresses that gap by combining the inverse-option valuation framework with affine pricing models that are appropriate for crypto derivatives and by testing the resulting framework on a large panel of Deribit BTC and ETH options. The contribution is fourfold. First, on the theoretical side, the thesis applies the numeraire-invariance logic of inverse valuation to options written on dated futures and formulates the corresponding premium-adjusted hedge ratio in a way that is directly usable in an empirical backtest. Second, on the modelling side, it embeds inverse option pricing in three nested specifications: the Black model as a constant-volatility benchmark, the Heston model as a stochastic-volatility benchmark, and the SVCJ model as a richer affine specification with correlated jumps in returns and variance. Third, on the empirical side, it calibrates these models repeatedly to Deribit option-chain snapshots for both BTC and ETH, thereby producing a time series of fitted parameters and out-of-sample pricing diagnostics. Fourth, on the risk-management side, it evaluates whether improvements in cross-sectional pricing accuracy translate into improvements in hedging performance once the hedge is implemented with the more liquid perpetual futures contract rather than the exact dated-futures underlying.

The central research question of the thesis can therefore be stated as follows: does a crypto-specific affine stochastic-volatility model with jumps materially improve the pricing and hedging of Deribit inverse options relative to standard benchmark models, once the inverse denomination of the contract and the practical use of perpetual futures in hedging are taken into account? This broad question can be decomposed into three more specific ones. First, can the SVCJ model fit the observed inverse option surface more accurately than Black and Heston? Second, does the premium-adjusted net delta implied by the inverse-option framework matter empirically for out-of-sample hedging performance? Third, do gains in pricing accuracy survive when the hedge is implemented as a delta (first-order) hedge with perpetual futures and is therefore exposed to basis risk relative to the option's dated-futures underlying?

The working hypothesis of the thesis has a corresponding three-part structure. The first part is that richer dynamics should matter for valuation: relative to Black and Heston, the SVCJ specification should provide a more accurate cross-sectional fit to observed Deribit inverse option prices, with the largest gains expected in the regions of the surface where jump risk and stochastic volatility are most relevant, namely the short end, the wings, and the more asymmetric parts of the smile. The second part is that inverse-contract-consistent hedging should matter for risk management: the premium-adjusted net delta should outperform the regular delta in out-of-sample hedging tests because it correctly accounts for the coin denomination of the option premium and payoff. The third part is that richer pricing models may also improve hedging performance, but this is a more delicate empirical question, since hedging is conducted with perpetual futures rather than with the exact dated-futures contract and is therefore subject to basis risk in addition to errors due to discrete rebalancing and model misspecification.

The empirical strategy is designed to test these claims directly. We first build the calibration framework from Deribit snapshots of BTC and ETH option chains, augmented with perpetual-futures data for the later hedging exercise. The raw option data are filtered to remove stale or illiquid quotes and to retain contracts with acceptable maturity and moneyness characteristics. Model parameters are then estimated separately for Black, Heston, and SVCJ by repeated cross-sectional calibration in coin-denominated prices, implemented as a weighted least-squares problem. The weights are designed to down-weight wide markets, while the optimization also incorporates temporal smoothing so that adjacent snapshots do not generate implausibly erratic parameter paths. Because this calibration must be repeated on dense option panels across many dates, pricing is implemented through characteristic-function methods and the Carr–Madan FFT procedure, which makes it computationally feasible to evaluate large numbers of strikes and maturities inside an optimization loop.

We then turn from valuation to hedging. Using the calibrated model parameters, the thesis constructs an out-of-sample interval-level hedging backtest on the test sample of options. For each observation time, the pricing model produces both the model-implied inverse option value and the corresponding regular delta; this is then converted into the premium-adjusted net delta implied by the inverse-option framework. The hedge is implemented using inverse perpetual futures, not the exact dated futures contract underlying the option, because the perpetual is the liquid instrument actually used in practice. The backtest therefore measures a realistic hedging problem rather than a perfect theoretical replication. Performance is evaluated using multiple diagnostics, including reductions in RMSE and MAE of the portfolio P&L relative to the unhedged position, reductions in variance, hit rates of absolute improvement, and left-tail risk measures such as the 5% quantile and expected shortfall of hedged interval P&L. This design allows the thesis to separate three effects: the general value of dynamic hedging, the specific value of using the inverse-option-consistent net delta, and the incremental value of moving from Black or Heston to a richer jump-diffusion model.

Taken together, the thesis studies inverse crypto options not as an exotic footnote to standard option theory, but as a natural market setting in which denomination, numeraire choice, affine pricing, and practical hedge implementation all matter. The problem is economically relevant because Deribit dominates listed crypto options activity, technically interesting because inverse contracts alter both valuation and Greeks, and empirically demanding because crypto option surfaces display features that call for stochastic volatility and jumps. The contribution of the thesis is to connect these strands in a single framework and to evaluate that framework using both pricing and hedging criteria. The remainder of the thesis is organised as follows:

1. Chapter 1 reviews the institutional setup of Deribit inverse options, the main theoretical results on inverse-option valuation and hedging, the evidence on stochastic volatility and jumps in cryptocurrency markets, and the literature gap that motivates the thesis.

2. Chapter 2 reviews the theoretical pricing and hedging framework, including the SVCJ dynamics under the futures numeraire, the derivation of the characteristic function, the Fourier pricing approach for inverse options, and the construction of regular and premium-adjusted hedge ratios.

3. Chapter 3 presents the data, cleaning and liquidity filters, cross-sectional calibration methodology, weighting and temporal smoothing scheme, and the out-of-sample pricing results for the Black, Heston, and SVCJ models.

4. Chapter 4 evaluates the out-of-sample hedging performance of the competing models, compares regular-delta and net-delta hedges implemented with perpetual fu-

tures, and analyses whether pricing improvements translate into economically meaningful reductions in hedging error.

The code used to implement the data collection and cleaning, pricing, calibration, and hedging routines, along with the collected Deribit BTC and ETH option chain snapshots, is available on GitHub.

1. Background & Literature Review

This chapter reviews the institutional setup of cryptocurrency inverse options on Deribit, the main theoretical approaches for pricing and hedging inverse options, empirical evidence that stochastic volatility and jumps are central for crypto option pricing, and the calibration of model parameters together with the Fourier/FFT pricing methodology that enables fast valuation of large option panels.

1.1 Crypto derivatives and Deribit inverse options

Crypto derivatives are primarily listed on specialised exchanges and are typically collateralised in crypto rather than fiat. Although direct options on cryptocurrencies denominated in fiat do exist, this market structure motivates the dominance of *inverse* contracts: instruments where the payoff and premium are denominated in the underlying coin, while the strike is typically quoted in USD. Since the empirical part of this thesis is based on Deribit option quotes, which is the most popular exchange for cryptocurrency options, we first clarify the mechanics and payoff conventions of inverse options.

Deribit lists European-style inverse options on BTC and ETH, with contract sizes equal to 1 BTC for BTC options and 1 ETH for ETH options. Option strikes are quoted in USD, while option prices (premia) and settlement amounts are denominated in the underlying coin. The underlying of a listed option is a termed futures contract (possibly synthetic) with the same expiry. At expiry, Deribit computes a *delivery price* as the time-weighted average of the Deribit index over a specified window, and then settles options in coin units based on intrinsic value divided by the delivery price (Deribit, 2025).

Let F_T denote the USD delivery price of the underlying futures contract at maturity T and K denote the USD strike. The (cash) USD intrinsic values are $(F_T - K)^+$ for calls and $(K - F_T)^+$ for puts. The corresponding inverse option payoffs in coin units are therefore

$$r_{\text{call}}(F_T) = \frac{(F_T - K)^+}{F_T}, \qquad r_{\text{put}}(F_T) = \frac{(K - F_T)^+}{F_T}. \tag{1.1.1}$$

1.2 Pricing and hedging inverse options

Inverse payoffs naturally arise when the numeraire is changed from USD cash (or a money-market account) to the underlying futures price. This viewpoint yields clean relationships between the price of a regular USD-settled option and the price of the corresponding inverse option. It also implies that the *delta* used for hedging inverse options differs from the textbook Black–Scholes/Black delta.

1.2.1 Lucic and Sepp (2024): numeraire invariance, valuation, and the "net delta"

The central inverse option pricing reference for this thesis is Lucic and Sepp (2024). They formalise inverse options using a martingale valuation approach and the *numeraire invariance principle* (Geman et al., 1995), which states that the value of a payoff does not depend on which asset you use as the numeraire, as long as prices are computed under the corresponding martingale measure. In their notation, F_t is the USD price of the underlying futures.

Their main result is a no-arbitrage identity linking direct (USD-settled) European option values and inverse (coin-settled) values. Let V_t^{USD} denote the USD value of the regular option and let V_t^{coin} denote the coin value of the inverse option. Then,

$$V_t^{\mathrm{USD}} = F_t \, V_t^{\mathrm{coin}}, \tag{1.2.1}$$

under suitable conditions ensuring the existence of both the USD and futures martingale measures (Lucic and Sepp, 2024). Equation (1.2.1) provides a direct route for inverse pricing: any model and numerical method that yields V_t^{USD} for a standard European option can be repurposed for inverse options by scaling with $1/F_t$.

A second key contribution of Lucic and Sepp (2024) is the model-independent delta relationship implied by (1.2.1). Differentiating with respect to F_t yields that the hedge ratio for the inverse option involves an adjustment by the option premium itself. The delta-hedging position in inverse futures contracts is given by

$$\Delta_t^{\mathrm{net}} = \Delta_t^{\mathrm{reg}} - \frac{V_t^{\mathrm{USD}}}{F_t} = \Delta_t^{\mathrm{reg}} - V_t^{\mathrm{coin}}, \tag{1.2.2}$$

where Δ_t^{reg} is the delta of the corresponding regular option, and Δ_t^{net} is the premium-adjusted or net delta of the inverse option.

Finally, Lucic and Sepp (2024) emphasise that delta hedging of listed inverse options is most naturally implemented using perpetual futures contracts. Although the underlying of a listed option is a termed futures contract with the same expiry, such term futures are often substantially less liquid than perpetual futures, or, in the case of synthetic futures underlyings, may not be tradeable at all. In contrast, perpetual futures are highly liquid and constitute the dominant delta-one instrument on major cryptocurrency exchanges and therefore serve as the primary vehicle for dynamic delta-hedging.

1.2.2 Alexander et al. (2023): inverse options as currency-pair/quanto contracts

An alternative interpretation of inverse options is to view them through an FX/currency-pair lens: holding BTC while measuring value in USD naturally resembles an exchange-rate setting. In Alexander et al. (2023), the authors provide a systematic exposition of direct and inverse crypto options, discuss market incompleteness in these markets, and frame inverse options alongside quanto structures that are relevant for fiat-based investors.

1.3 Why stochastic volatility and jumps matter in crypto option pricing

Empirically, cryptocurrency return distributions exhibit heavy tails, sharp discontinuities, and strong time-variation in volatility. In option markets, these features manifest as pronounced implied volatility smiles/skews and term structures that are difficult to reconcile with constant-volatility diffusion structures. This section focuses on empirical evidence supporting the use of stochastic volatility and jump components when modelling crypto option prices.

1.3.1 Scaillet et al. (2020): high-frequency evidence of jumps

Scaillet et al. (2020) study high-frequency Bitcoin data and document economically significant jump behaviour. While their focus is not on option pricing, their results provide foundational empirical motivation for jump components in option pricing models: if jumps are a structural feature of the underlying, option-implied distributions and tails are naturally affected. This type of evidence supports modelling choices that allow discontinuities rather than relying on pure diffusion volatility to match option price surfaces.

1.3.2 Chen et al. (2024): option-implied evidence linking IV slopes to jumps

Chen et al. (2024) provide option-market evidence that connects the shape of the implied volatility surface to jump risk. They develop a theoretical relation between implied volatility slopes and positive/negative price jumps, and empirically study this mechanism using tick-by-tick Bitcoin options data from Deribit. Their results support the view that jump components are not merely a convenient modelling choice but are reflected in the cross-sectional structure of implied volatility.

1.3.3 Chaim and Laurini (2018) and Shen et al. (2020): evidence of jumps in volatility

Evidence for jumps in crypto markets extends beyond discontinuities in returns: volatility itself also exhibits abrupt, jump-like behavior. Chaim and Laurini (2018) model Bitcoin with stochastic volatility and jumps and find that volatility jumps are economically meaningful; in particular, they report that volatility shocks associated with jumps have persistent effects, consistent with the idea that volatility dynamics are not well captured by purely continuous diffusions.

Complementary high-frequency evidence comes from Shen et al. (2020), who study Bitcoin volatility forecasting using realized volatility decompositions that explicitly separate continuous and jump components. Their results indicate that jump-related components of realized variance are important for explaining and forecasting volatility, reinforcing the idea that volatility contains discontinuous movements that are relevant for option pricing.

1.3.4 Hou et al. (2020): SVCJ modelling and evidence from crypto options

The core modelling reference for this thesis is Hou et al. (2020), who propose pricing cryptocurrency options with a stochastic volatility model that includes correlated jumps in both the price and volatility processes (SVCJ model). Their motivation is that price jumps and volatility jumps co-occur in crypto markets and can materially affect option prices, especially for short maturities and far-from-the-money strikes.

A key feature of their specification is a joint jump structure that allows dependence between return jumps and volatility jumps. In simplified form (suppressing some risk-neutral drift details), the SVCJ dynamics are

$$d\log F_t = \big(\,\cdot\,\big)\,dt + \sqrt{V_t}\,dW_t^S + Z_t^y\,dN_t, \tag{1.3.1}$$

$$dV_t = \kappa(\theta - V_t)\,dt + \sigma_v\sqrt{V_t}\,dW_t^V + Z_t^v\,dN_t, \tag{1.3.2}$$

where N_t is a Poisson process with intensity λ, and (W^S, W^V) are correlated Brownian motions (Hou et al., 2020). The jump sizes are specified to allow co-jumps and dependence through a conditional distribution:

$$Z_t^v \sim \text{Exponential}(\ell_v), \qquad Z_t^y \,\big|\, Z_t^v \sim \mathcal{N}\big(\ell_y + q_j Z_t^v,\ r_y^2\big)\,, \tag{1.3.3}$$

so that the price jump distribution depends on the volatility jump size through q_j. Empirically, Hou et al. (2020) report that incorporating this structure is important for fitting crypto option prices, and they document economically meaningful co-jump features in their estimation results.

For the purposes of this thesis, the significance of Hou et al. (2020) is twofold. First, it provides a concrete affine SV-with-jumps specification that is empirically supported in cryptocurrency data. Second, it offers a modelling backbone that can be combined with the inverse-option valuation identity in Lucic and Sepp (2024), thereby enabling inverse option pricing under SVCJ dynamics.

1.4 Research gap and contribution

Sections 1.1–1.3 highlight that (i) the most liquid listed crypto options are inverse contracts, (ii) inverse pricing and delta conventions are theoretically well understood under numeraire invariance, and (iii) stochastic volatility and jumps are empirically important for crypto option surfaces. However, these strands have not yet been fully integrated in a single empirical study focused on inverse options.

Inverse option valuation: Lucic and Sepp (2024) provide the key valuation identity (1.2.1) and the premium-adjusted delta formula (1.2.2), together with an empirical study of systematic strategies and the role of perpetual futures for hedging. Their focus, however, is not on developing a crypto-specific stochastic volatility with jumps model for inverse option prices. Similarly, Alexander et al. (2023) clarify inverse and quanto structures and relate them to FX pricing intuition, but do not develop an empirical pricing and calibration framework for inverse option surfaces that includes stochastic volatility and jumps.

Stochastic volatility and jump models for crypto option pricing: Empirical and option-market studies provide strong evidence that stochastic volatility and jumps in returns and volatility matter in crypto contexts (Scaillet et al., 2020; Chen et al., 2024; Chaim and Laurini, 2018; Shen et al., 2020). In particular, Hou et al. (2020) develop and empirically support an SVCJ framework for pricing cryptocurrency options. Yet, their analysis is primarily aligned with "direct" (USD-denominated) options on the spot asset as the underlying, and does not explicitly target the inverse payoff structure of options on termed futures listed on Deribit.

This thesis aims to fill the above gaps by combining the inverse-option valuation framework with a crypto-specific affine stochastic volatility model with jumps, calibrating the SVCJ model to Deribit options data, and evaluating it against a set of baseline models in terms of pricing and hedging performance. The workflow is the following:

- **Theory:** Use the numeraire invariance identity of Lucic and Sepp (2024) to price inverse options on futures via standard (USD-like) option prices, and derive the corresponding hedging delta using the premium-adjusted formula (1.2.2).
- **Model:** Adopt the SVCJ model in Hou et al. (2020) (equation (1.3.3)) to price inverse options on futures.
- **Empirics:** Calibrate the competing models (Black, Heston, SVCJ) to Deribit BTC and ETH inverse option panels and produce time series of parameter estimates by repeating calibration on successive option-chain snapshots.
- **Evaluation:** Compare models using out-of-sample pricing errors and, additionally, assess risk-management relevance via delta-hedging performance.

The working hypotheses of the paper can be formulated as follows:
A crypto-specific affine stochastic volatility model with correlated jumps in the return and volatility processes (SVCJ) improves (i) cross-sectional fit to observed market prices, (ii) delta-hedging performance via the use of the net-delta instead of the regular delta as the hedge ratio, and (iii) the richer model dynamics of SVCJ translate to better delta-hedging performance for Deribit inverse options relative to Black and Heston benchmarks.

The corresponding testable implications of these hypotheses are

- lower out-of-sample pricing error for the SVCJ model compared to Black and Heston baselines,
- lower hedging error when using the premium-adjusted net-delta as the hedge ratio, and
- lower hedging error when using the SVCJ model to compute the hedge ratio compared to the baselines.

Before turning to the model, the next two sections briefly introduce the calibration and Fourier-pricing machinery used throughout the empirical chapters.

1.5 Calibration overview: cross-sectional fitting by MSE minimization

In applied option pricing, model parameters are typically estimated by *cross-sectional* calibration: at each observation time t one fits a parameter vector Θ_t to the set of contemporaneous option quotes across strikes and maturities. The calibration objective is commonly a mean squared error (MSE) criterion based on either prices or implied volatilities.

Let $\{(K_i, T_i)\}_{i=1}^{N_t}$ denote the observed option contracts at time t, and let $C_{t,i}^{\text{mkt}}$ denote market prices (or implied vols). A standard calibration problem is

$$\widehat{\Theta}_t \in \arg\min_{\Theta \in \mathbb{X}} \sum_{i=1}^{N_t} w_{t,i} \Big(C^{\text{model}}(t; K_i, T_i, \Theta) - C_{t,i}^{\text{mkt}} \Big)^2, \tag{1.5.1}$$

where $w_{t,i} \geq 0$ are weights (e.g., uniform weights, or weights that down-weight illiquid options), and $\mathbb{X}$ is the admissible parameter set (e.g., enforcing positivity of variance processes, Feller-type constraints, stationarity restrictions, etc.). In crypto option applications, the large number of options per snapshot and the need to repeat (1.5.1) over time makes fast pricing methods essential (Madan et al., 2019).

In this thesis, "time-varying calibration results" correspond to the sequence $\{\widehat{\Theta}_{t_j}\}_{j=1}^{J}$ obtained by applying (1.5.1) at each snapshot time t_j, separately for BTC and ETH panels. This produces parameter time series that can be analysed for stability, regime shifts, and their relation to market conditions.

1.6 Fourier-based pricing: characteristic functions and FFT

Cross-sectional calibration requires pricing hundreds to thousands of options repeatedly. For affine stochastic volatility models, a standard approach is to price European options via the characteristic function of the log-price under the relevant pricing measure.

If $X_T = \log F_T$, and if the model yields a closed-form characteristic function $\phi(u; t, T) = \mathbb{E}[\exp(iuX_T) \mid \mathcal{F}_t]$, then option prices can be obtained using Fourier inversion techniques. In particular, the FFT method of Carr and Madan (1999) converts option valuation across a grid of strikes into fast convolution-type computations, making it highly suitable for calibration loops. The reason affine models are especially convenient is that their characteristic functions are exponential-affine in the state variables, as exemplified by the Heston model (Heston, 1993) and formalised for affine jump diffusion processes in Duffie et al. (2000).

For inverse options, the valuation identity (1.2.1) implies a practical workflow:

1. compute the corresponding regular (USD-like) European option value V_t^{USD} under the chosen affine model using characteristic functions and FFT;
2. obtain the inverse option value by $V_t^{\mathrm{coin}} = V_t^{\mathrm{USD}}/F_t$.

This is the computational strategy adopted in Chapter 2.

2. Pricing and Hedging Inverse Options under the SVCJ Model

This chapter develops a pricing and hedging framework for cryptocurrency inverse European options traded on Deribit, using the stochastic volatility model with correlated jumps in returns and volatility (SVCJ). We proceed in five steps: (i) specify the SVCJ dynamics as in Hou et al. (2020) and adapt it to options written on termed futures; (ii) derive the characteristic function of the model from first principles; (iii) price inverse options under the futures numeraire following the logic of Lucic and Sepp (2024); (iv) derive model deltas and the premium-adjusted net delta used for inverse option hedging; (v) show how an inverse option on a term futures contract can be hedged using the more liquid inverse perpetual futures and how to evaluate hedging performance via hedged P&L.

Throughout, we focus on European inverse options on a dated futures contract with the same expiry as the option, which is the standard convention on Deribit.

2.1 The SVCJ model for term-futures underlyings

2.1.1 Baseline SVCJ specification (Hou et al., 2020)

Let $F_t \equiv F_t^{(T)}$ denote the *term futures* price (quoted in USD per coin) for maturity T, observed at time $t \in [0, T]$. Define the log-futures price

$$X_t := \log F_t. \tag{2.1.1}$$

Following Hou et al. (2020), we consider an SVCJ model in which both the log-price and the variance process jump together through a common Poisson process. Let N_t be a Poisson process with constant intensity $\lambda > 0$ and let (W_{1t}, W_{2t}) be two Brownian motions with instantaneous correlation $\rho \in (-1, 1)$:

$$\operatorname{corr}(\mathrm{d}W_{1t}, \mathrm{d}W_{2t}) = \rho. \tag{2.1.2}$$

The SVCJ dynamics under an *arbitrary pricing measure* take the form

$$\boxed{\begin{aligned} \mathrm{d}X_t &= b_t\,\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_{1t} + Z^y\,\mathrm{d}N_t, \\ \mathrm{d}V_t &= \kappa(\theta - V_t)\,\mathrm{d}t + \sigma_v\sqrt{V_t}\,\mathrm{d}W_{2t} + Z^v\,\mathrm{d}N_t, \end{aligned}} \qquad \begin{aligned} &(2.1.3) \\ &(2.1.4) \end{aligned}$$

with parameters $\kappa > 0$ (mean reversion), $\theta > 0$ (long-run variance), $\sigma_v > 0$ (vol-of-vol), and $V_t \geq 0$. The drift b_t depends on the chosen pricing measure, and is derived below.

The jump sizes (Z^y, Z^v) are specified as in Hou et al. (2020): the volatility jump Z^v is exponentially distributed, and the return jump Z^y is Gaussian conditional on Z^v:

$$Z^v \sim \mathrm{Exp}(\text{mean } \ell_v), \qquad \ell_v > 0, \tag{2.1.5}$$

$$Z^y \mid Z^v \sim \mathcal{N}\big(\ell_y + \rho_j Z^v,\, \sigma_y^2\big), \qquad \sigma_y > 0. \tag{2.1.6}$$

Here $\ell_y \in \mathbb{R}$ controls the average log-price jump, σ_y controls the variance of log-price jumps, and $\rho_j \in \mathbb{R}$ controls the dependence between price-jump and volatility-jump sizes.

2.1.2 SVCJ model under the futures price numeraire

For options written on a futures contract with the same maturity T and with deterministic discounting, it is natural to work under a measure under which F_t is a martingale.[1]

We therefore choose the drift b_t in (2.1.3) so that the term-futures price

$$F_t := e^{X_t}$$

is a (local) martingale under the chosen pricing measure. Since F_t may jump when X_t jumps, it is convenient to use the standard left-limit notation

$$F_{t^-} := \lim_{s \uparrow t} F_s,$$

i.e., the value of the process immediately before time t (equivalently, F_{t^-} is the pre-jump value, while F_t is the post-jump value). The martingale condition can then be written as

$$\mathbb{E}\left[\frac{\mathrm{d}F_t}{F_{t^-}} \,\middle|\, \mathcal{F}_{t^-}\right] = 0. \tag{2.1.7}$$

Theorem (Itô's formula for jump diffusions, Øksendal and Sulem (2007))**.** *Let $(Y_t)_{t\geq 0}$ be a one-dimensional jump–diffusion process of the form*

$$\mathrm{d}Y_t = \mathrm{d}Y_t^{(c)} + \Delta Y_t \,\mathrm{d}N_t, \qquad \mathrm{d}Y_t^{(c)} := a_t \,\mathrm{d}t + b_t \,\mathrm{d}W_t, \tag{2.1.8}$$

where $(W_t)_{t\geq 0}$ is a Brownian motion, $(N_t)_{t\geq 0}$ is a Poisson process, and ΔY_t denotes the jump size at a jump time. Let $f : \mathbb{R} \to \mathbb{R}$ be twice continuously differentiable. Then,

$$\mathrm{d}f(Y_t) = f'(Y_{t^-})\,\mathrm{d}Y_t^{(c)} + \frac{1}{2} f''(Y_{t^-})\,\mathrm{d}[Y^{(c)}]_t + \Big(f(Y_{t^-} + \Delta Y_t) - f(Y_{t^-})\Big)\,\mathrm{d}N_t, \tag{2.1.9}$$

where $Y_{t^-} = \lim_{s\uparrow t} Y_s$, and $\mathrm{d}[Y^{(c)}]_t = b_t^2 \,\mathrm{d}t$ is the quadratic variation of the continuous part of the process.

[1] For crypto inverse options on Deribit, discounting is assumed to be negligible (Lucic and Sepp, 2024).

Now, apply Itô's formula for jump diffusions to $F_t = f(X_t) = e^{X_t}$. Separating the continuous and jump parts of X_t:

$$\mathrm{d}X_t = \mathrm{d}X_t^{(c)} + Z^y\,\mathrm{d}N_t, \qquad \mathrm{d}X_t^{(c)} := b_t\,\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_{1t}.$$

The continuous quadratic variation is

$$\mathrm{d}[X^{(c)}]_t = (\mathrm{d}X_t^{(c)})^2 = V_t\,\mathrm{d}t.$$

Since $f'(x) = f''(x) = e^x$, Itô's formula gives

$$\begin{aligned}
\mathrm{d}F_t &= f'(X_{t-})\,\mathrm{d}X_t^{(c)} + \frac{1}{2}f''(X_{t-})\,\mathrm{d}[X^{(c)}]_t + \Big(f(X_{t-} + Z^y) - f(X_{t-})\Big)\,\mathrm{d}N_t \\
&= F_{t-}\Big(b_t\,\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_{1t}\Big) + \frac{1}{2}F_{t-}V_t\,\mathrm{d}t + F_{t-}\big(e^{Z^y} - 1\big)\,\mathrm{d}N_t \\
&= F_{t-}\left(b_t + \frac{1}{2}V_t\right)\mathrm{d}t + F_{t-}\sqrt{V_t}\,\mathrm{d}W_{1t} + F_{t-}\big(e^{Z^y} - 1\big)\,\mathrm{d}N_t. \qquad (2.1.10)
\end{aligned}$$

Dividing by F_{t-} yields

$$\frac{\mathrm{d}F_t}{F_{t-}} = \left(b_t + \frac{1}{2}V_t\right)\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_{1t} + \big(e^{Z^y} - 1\big)\,\mathrm{d}N_t. \qquad (2.1.11)$$

which is the expression used to impose the martingale condition (2.1.7).

Taking conditional expectations in (2.1.11), we compute

$$\mathbb{E}\left[\frac{\mathrm{d}F_t}{F_{t-}}\,\middle|\,\mathcal{F}_{t-}\right] = \mathbb{E}\left[\left(b_t + \frac{1}{2}V_t\right)\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_{1t} + \big(e^{Z^y} - 1\big)\,\mathrm{d}N_t\,\middle|\,\mathcal{F}_{t-}\right]. \qquad (2.1.12)$$

Since $\mathrm{d}W_{1t}$ has zero conditional mean and is independent of $\mathcal{F}_{t-}$, and since $\mathrm{d}N_t$ is independent of $\mathcal{F}_{t-}$ with $\mathbb{E}[\mathrm{d}N_t \mid \mathcal{F}_{t-}] = \lambda\,\mathrm{d}t$, this reduces to

$$\begin{aligned}
\mathbb{E}\left[\frac{\mathrm{d}F_t}{F_{t-}}\,\middle|\,\mathcal{F}_{t-}\right] &= \left(b_t + \frac{1}{2}V_t\right)\mathrm{d}t + \mathbb{E}\big[e^{Z^y} - 1\big]\;\mathbb{E}[\mathrm{d}N_t \mid \mathcal{F}_{t-}] \\
&= \left[\left(b_t + \frac{1}{2}V_t\right) + \lambda\,\mathbb{E}\big[e^{Z^y} - 1\big]\right]\mathrm{d}t. \qquad (2.1.13)
\end{aligned}$$

Imposing the martingale condition $\mathbb{E}[\mathrm{d}F_t/F_{t-} \mid \mathcal{F}_{t-}] = 0$ therefore yields

$$\left(b_t + \frac{1}{2}V_t\right) + \lambda\,\mathbb{E}\big[e^{Z^y} - 1\big] = 0, \qquad (2.1.14)$$

and hence the drift of the log-futures process must be chosen as

$$\boxed{b_t = -\frac{1}{2}V_t - \lambda\kappa_F, \qquad \kappa_F := \mathbb{E}\big[e^{Z^y} - 1\big]\,.} \qquad (2.1.15)$$

Under the Hou-type jump specification (2.1.5)–(2.1.6), κ_F is available in closed form:

$$\mathbb{E}\big[e^{Z^y} \mid Z^v\big] = \exp\!\left(\ell_y + \rho_j Z^v + \frac{1}{2}\sigma_y^2\right), \qquad (2.1.16)$$

$$\mathbb{E}\big[e^{Z^y}\big] = \exp\!\left(\ell_y + \frac{1}{2}\sigma_y^2\right)\,\mathbb{E}\big[e^{\rho_j Z^v}\big]\,. \qquad (2.1.17)$$

If Z^v is exponential with mean ℓ_v, then for any complex s with $\mathrm{Re}(s) < 1/\ell_v$,

$$\mathbb{E}[e^{sZ^v}] = \frac{1}{1-\ell_v s}. \tag{2.1.18}$$

Therefore, provided $1 - \ell_v \rho_j > 0$ (when $\rho_j > 0$),

$$\boxed{\kappa_F = \exp\left(\ell_y + \frac{1}{2}\sigma_y^2\right) \frac{1}{1-\ell_v \rho_j} - 1.} \tag{2.1.19}$$

Then, the SVCJ dynamics under the futures price numeraire are as follows:

SVCJ dynamics under the futures price numeraire

$$\mathrm{d}X_t = \left(-\frac{1}{2}V_t - \lambda \kappa_F\right) \mathrm{d}t + \sqrt{V_t}\, \mathrm{d}W_{1t} + Z^y\, \mathrm{d}N_t, \tag{2.1.20}$$

$$\mathrm{d}V_t = \kappa(\theta - V_t)\, \mathrm{d}t + \sigma_v \sqrt{V_t}\, \mathrm{d}W_{2t} + Z^v\, \mathrm{d}N_t, \tag{2.1.21}$$

with $\mathrm{corr}(\mathrm{d}W_{1t}, \mathrm{d}W_{2t}) = \rho$, where N_t is a Poisson process with intensity λ, and the jump sizes satisfy

$$Z^v \sim \text{Exponential}(\text{mean } \ell_v), \tag{2.1.22}$$

$$Z^y \mid Z^v \sim \mathcal{N}\left(\ell_y + \rho_j Z^v,\, \sigma_y^2\right). \tag{2.1.23}$$

The jump compensator $\kappa_F = \mathbb{E}[e^{Z^y} - 1]$ is given in closed form by

$$\kappa_F = \exp\left(\ell_y + \frac{1}{2}\sigma_y^2\right) \frac{1}{1-\ell_v \rho_j} - 1, \qquad 1 - \ell_v \rho_j > 0. \tag{2.1.24}$$

The parameters of the model are:

$$\Theta_{\mathrm{SVCJ}} = \left(\kappa, \theta, \sigma_v, \rho, \lambda, \ell_y, \sigma_y, \ell_v, \rho_j\right). \tag{2.1.25}$$

To ensure non-negativity of the variance process, we additionally impose the standard Feller condition

$$2\kappa\theta \geq \sigma_v^2, \tag{2.1.26}$$

which is sufficient (though not necessary) for V_t to remain strictly positive (Heston, 1993). Additional admissibility conditions include $\lambda > 0$, $\ell_v > 0$, and $1 - \ell_v \rho_j > 0$, ensuring finite jump moments and well-defined characteristic functions.

2.2 Deriving the characteristic function for the SVCJ model

This section derives the characteristic function of $X_T = \log F_T$ conditional on (X_t, V_t) in the SVCJ model under the martingale drift choice (2.1.15). The derivation follows the affine-jump-diffusion logic underlying SVCJ-type models (Duffie et al., 2000; Hou et al., 2020).

2.2.1 Martingale approach to the characteristic function

We derive the characteristic function using a martingale argument based on Itô's formula for jump diffusions.

Fix $u \in \mathbb{C}$ and define the *conditional characteristic function*

$$\boxed{\phi(u;t,T) := \mathbb{E}\left[e^{iuX_T} \mid X_t = x,\ V_t = v\right].} \tag{2.2.1}$$

Since (X_t, V_t) solves a system of stochastic differential equations whose coefficients depend only on the current state and are driven by Brownian motions and a Poisson process with independent increments, the process (X_t, V_t) is a time-homogeneous Markov process (Øksendal and Sulem, 2007). Then, $\phi(u;t,T)$ depends on the current state only through (t, x, v). Our objective is to determine ϕ explicitly.

Motivated by the affine structure of the SVCJ model, we conjecture that the characteristic function admits an exponential-affine representation of the form

$$\phi(u;t,T) = \exp\big(A(\tau;u) + B(\tau;u)V_t + iuX_t\big), \qquad \tau := T - t, \tag{2.2.2}$$

with terminal conditions

$$A(0;u) = 0, \qquad B(0;u) = 0,$$

so that $\phi(u;T,T) = e^{iuX_T}$ (Duffie et al., 2000).

Define the process

$$\Phi_t := \exp\big(A(\tau;u) + B(\tau;u)V_t + iuX_t\big). \tag{2.2.3}$$

If the functions $A(\cdot;u)$ and $B(\cdot;u)$ are chosen such that $(\Phi_t)_{t\in[0,T]}$ is a martingale, then by construction

$$\Phi_t = \mathbb{E}[\Phi_T \mid \mathcal{F}_t] = \mathbb{E}\left[e^{iuX_T} \mid \mathcal{F}_t\right],$$

and hence $\Phi_t = \phi(u;t,T)$ is the characteristic function.

2.2.2 Application of Itô's formula to Φ_t

We derive the *Riccati equations* for $A(\tau; u)$ and $B(\tau; u)$, i.e., the ODEs that can be solved to get the characteristic function, by applying Itô's formula for jump diffusions to the exponential-affine process

$$\Phi_t := \exp\big(A(\tau; u) + B(\tau; u)V_t + iuX_t\big), \qquad \tau := T - t, \tag{2.2.4}$$

where $u \in \mathbb{C}$ is fixed. The functions $A(\tau; u)$ and $B(\tau; u)$ are deterministic in τ and will be chosen so that $(\Phi_t)_{t\in[0,T]}$ is a martingale. Note that $\tau = T - t$ implies $\mathrm{d}\tau = -\mathrm{d}t$.

Theorem (Itô's formula for two-dimensional jump–diffusions, (Øksendal and Sulem, 2007))**.** *Let $(X_t, V_t)_{t\geq 0}$ be a two-dimensional jump–diffusion of the form*

$$\mathrm{d}X_t = \mathrm{d}X_t^{(c)} + Z^y \,\mathrm{d}N_t, \tag{2.2.5}$$

$$\mathrm{d}V_t = \mathrm{d}V_t^{(c)} + Z^v \,\mathrm{d}N_t, \tag{2.2.6}$$

where $(X_t^{(c)}, V_t^{(c)})$ denotes the continuous part, N_t is a Poisson process, and (Z^y, Z^v) denotes the jump sizes (possibly random, and independent of $\mathcal{F}_{t-}$). Let $\Phi : [0,\infty) \times \mathbb{R} \times \mathbb{R}_+ \to \mathbb{R}$ be once continuously differentiable in t and twice continuously differentiable in (x, v). Define $\Phi_t := \Phi(t, X_t, V_t)$ and denote left limits by $X_{t^-} := \lim_{s\uparrow t} X_s$ and $V_{t^-} := \lim_{s\uparrow t} V_s$. Then Φ_t admits the decomposition

$$\mathrm{d}\Phi_t = \mathrm{d}\Phi_t^{(c)} + \mathrm{d}\Phi_t^{(J)}, \tag{2.2.7}$$

where the continuous part is

$$\begin{aligned}
\mathrm{d}\Phi_t^{(c)} = {} & \frac{\partial \Phi}{\partial t}(t, X_{t^-}, V_{t^-})\,\mathrm{d}t + \frac{\partial \Phi}{\partial x}(t, X_{t^-}, V_{t^-})\,\mathrm{d}X_t^{(c)} + \frac{\partial \Phi}{\partial v}(t, X_{t^-}, V_{t^-})\,\mathrm{d}V_t^{(c)} \\
& + \frac{1}{2}\frac{\partial^2 \Phi}{\partial x^2}(t, X_{t^-}, V_{t^-})\,\big(\mathrm{d}X_t^{(c)}\big)^2 + \frac{1}{2}\frac{\partial^2 \Phi}{\partial v^2}(t, X_{t^-}, V_{t^-})\,\big(\mathrm{d}V_t^{(c)}\big)^2 \\
& + \frac{\partial^2 \Phi}{\partial x\,\partial v}(t, X_{t^-}, V_{t^-})\,\mathrm{d}X_t^{(c)}\,\mathrm{d}V_t^{(c)},
\end{aligned} \tag{2.2.8}$$

and the jump part is

$$\mathrm{d}\Phi_t^{(J)} = \Big(\Phi(t, X_{t^-} + Z^y, V_{t^-} + Z^v) - \Phi(t, X_{t^-}, V_{t^-})\Big)\,\mathrm{d}N_t. \tag{2.2.9}$$

Now, applying Itô's formula for two-dimensional jump–diffusions to the process Φ_t.

Step 1: derivatives of $\Phi(t, x, v)$. For $\Phi(t, x, v) = \exp(A(\tau; u) + B(\tau; u)v + iux)$ with $\tau = T - t$, the spatial derivatives are

$$\frac{\partial \Phi}{\partial x} = iu\,\Phi, \qquad \frac{\partial \Phi}{\partial v} = B(\tau; u)\,\Phi,$$

$$\frac{\partial^2\Phi}{\partial x^2} = -u^2\,\Phi, \qquad \frac{\partial^2\Phi}{\partial v^2} = B(\tau;u)^2\,\Phi, \qquad \frac{\partial^2\Phi}{\partial x\,\partial v} = iu\,B(\tau;u)\,\Phi.$$

The time derivative must account for $\tau = T - t$:

$$\frac{\partial\Phi}{\partial t} = \Phi\left(\frac{\partial}{\partial t}\big(A(\tau;u) + B(\tau;u)v\big)\right) = \Phi\left(-A_\tau(\tau;u) - B_\tau(\tau;u)\,v\right),$$

where $A_\tau(\tau;u) := \partial A(\tau;u)/\partial\tau$ and $B_\tau(\tau;u) := \partial B(\tau;u)/\partial\tau$.

Step 2: dynamics and quadratic variations. Under the futures price numeraire, split each state variable into its continuous and jump parts:

$$\mathrm{d}X_t = \mathrm{d}X_t^{(c)} + Z^y\,\mathrm{d}N_t, \qquad \mathrm{d}X_t^{(c)} = \left(-\frac{1}{2}V_t - \lambda\kappa_F\right)\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_{1t}, \tag{2.2.10}$$

$$\mathrm{d}V_t = \mathrm{d}V_t^{(c)} + Z^v\,\mathrm{d}N_t, \qquad \mathrm{d}V_t^{(c)} = \kappa(\theta - V_t)\mathrm{d}t + \sigma_v\sqrt{V_t}\,\mathrm{d}W_{2t}, \tag{2.2.11}$$

with $\mathrm{corr}(\mathrm{d}W_{1t}, \mathrm{d}W_{2t}) = \rho$. The *continuous* quadratic covariations are

$$(\mathrm{d}X_t^{(c)})^2 = V_t\,\mathrm{d}t, \qquad (\mathrm{d}V_t^{(c)})^2 = \sigma_v^2 V_t\,\mathrm{d}t, \qquad \mathrm{d}X_t^{(c)}\,\mathrm{d}V_t^{(c)} = \rho\sigma_v V_t\,\mathrm{d}t.$$

Step 3: continuous part of $\mathrm{d}\Phi_t$. Substituting the derivatives into (2.2.8) and using the quadratic variations yields

$$\begin{aligned}\mathrm{d}\Phi_t^{(c)} = \Phi_t\Bigg\{\Big[&- A_\tau(\tau;u) - B_\tau(\tau;u)V_t + iu\Big(-\tfrac{1}{2}V_t - \lambda\kappa_F\Big) + B(\tau;u)\kappa(\theta - V_t)\\ &- \tfrac{1}{2}u^2 V_t + \tfrac{1}{2}\sigma_v^2 B(\tau;u)^2 V_t + \rho\sigma_v iu\,B(\tau;u)V_t\Big]\mathrm{d}t\\ &+ iu\sqrt{V_t}\,\mathrm{d}W_{1t} + \sigma_v B(\tau;u)\sqrt{V_t}\,\mathrm{d}W_{2t}\Bigg\}.\end{aligned} \tag{2.2.12}$$

Step 4: jump part of $\mathrm{d}\Phi_t$. At a jump time, (X_t, V_t) moves from (X_{t^-}, V_{t^-}) to $(X_{t^-} + Z^y, V_{t^-} + Z^v)$, so

$$\Phi(t, X_{t^-} + Z^y, V_{t^-} + Z^v) - \Phi(t, X_{t^-}, V_{t^-}) = \Phi_{t^-}\left(e^{iuZ^y + B(\tau;u)Z^v} - 1\right), \tag{2.2.13}$$

and therefore

$$\mathrm{d}\Phi_t^{(J)} = \Phi_{t^-}\left(e^{iuZ^y + B(\tau;u)Z^v} - 1\right)\mathrm{d}N_t. \tag{2.2.14}$$

Step 5: full dynamics and predictable drift. Combining (2.2.12) and (2.2.14), it is convenient to rewrite the jump term using the *compensated Poisson process*

$$\widetilde{N}_t := N_t - \lambda t, \qquad \text{so that} \qquad \mathrm{d}N_t = \mathrm{d}\widetilde{N}_t + \lambda\,\mathrm{d}t,$$

and $\mathbb{E}[\mathrm{d}\widetilde{N}_t \mid \mathcal{F}_{t-}] = 0$. Substituting this decomposition into the jump contribution yields the semimartingale dynamics

$$\mathrm{d}\Phi_t = \Phi_{t-}\Big(\widetilde{\mathcal{D}}_t\,\mathrm{d}t + iu\sqrt{V_t}\,\mathrm{d}W_{1t} + \sigma_v B(\tau;u)\sqrt{V_t}\,\mathrm{d}W_{2t}\Big) + \Phi_{t-}\,\big(e^{iuZ^y + B(\tau;u)Z^v} - 1\big)\,\mathrm{d}\widetilde{N}_t, \tag{2.2.15}$$

where the *predictable drift coefficient* $\widetilde{\mathcal{D}}_t$ is

$$\begin{aligned}\widetilde{\mathcal{D}}_t = {} & -A_\tau(\tau;u) - B_\tau(\tau;u)V_t + \kappa(\theta - V_t)B(\tau;u) - \frac{1}{2}(u^2 + iu)V_t + \frac{1}{2}\sigma_v^2 B(\tau;u)^2 V_t \\ & + \rho\sigma_v iu\, B(\tau;u)V_t - iu\lambda\kappa_F + \lambda\Big(M(u, B(\tau;u)) - 1\Big),\end{aligned} \tag{2.2.16}$$

with the *jump transform*

$$M(u,B) := \mathbb{E}\big[e^{iuZ^y + BZ^v}\big]\,. \tag{2.2.17}$$

Equation (2.2.15) separates $\mathrm{d}\Phi_t$ into a predictable drift term, a continuous local martingale part driven by (W_{1t}, W_{2t}), and a purely discontinuous martingale part driven by the compensated jump process $\widetilde{N}_t$.

Step 6: drift cancellation and Riccati equations. Requiring (Φ_t) to be a martingale implies $\widetilde{\mathcal{D}}_t \equiv 0$ for all t and all states. Since (2.2.16) is affine in V_t, we set separately the coefficient of V_t and the constant term to zero. Some algebra yields the *Riccati system*

$$\boxed{\begin{aligned} B_\tau(\tau;u) &= \frac{1}{2}\sigma_v^2 B(\tau;u)^2 + \big(\rho\sigma_v iu - \kappa\big)B(\tau;u) - \frac{1}{2}\big(u^2 + iu\big), \qquad B(0;u) = 0, \\ A_\tau(\tau;u) &= \kappa\theta\, B(\tau;u) + \lambda\Big(M(u, B(\tau;u)) - 1 - iu\kappa_F\Big), \qquad A(0;u) = 0. \end{aligned}} \quad \begin{aligned}(2.2.18)\\(2.2.19)\end{aligned}$$

Solving (2.2.18)–(2.2.19) yields the conditional characteristic function

$$\phi(u;t,T) = \mathbb{E}\big[e^{iuX_T} \mid X_t, V_t\big] = \exp\big(A(\tau;u) + B(\tau;u)V_t + iuX_t\big), \qquad \tau = T - t.$$

Step 7: closed form of the jump transform Under (2.1.5)–(2.1.6), conditional on $Z^v = z$,

$$\mathbb{E}\big[e^{iuZ^y} \mid Z^v = z\big] = \exp\bigg(iu(\ell_y + \rho_j z) - \frac{1}{2}u^2\sigma_y^2\bigg)\,. \tag{2.2.20}$$

Thus,

$$M(u,B) = \mathbb{E}\bigg[\exp\bigg(iu(\ell_y + \rho_j Z^v) - \frac{1}{2}u^2\sigma_y^2\bigg)\,e^{BZ^v}\bigg] = \exp\bigg(iu\ell_y - \frac{1}{2}u^2\sigma_y^2\bigg)\,\mathbb{E}\big[e^{(B + iu\rho_j)Z^v}\big]\,. \tag{2.2.21}$$

If Z^v is exponential with mean ℓ_v, then

$$\mathbb{E}\big[e^{(B+iu\rho_j)Z^v}\big] = \frac{1}{1 - \ell_v(B + iu\rho_j)}, \qquad \mathrm{Re}(B + iu\rho_j) < \frac{1}{\ell_v}, \tag{2.2.22}$$

so the jump transform is

$$\boxed{M(u,B) = \exp\bigg(iu\ell_y - \frac{1}{2}u^2\sigma_y^2\bigg)\,\frac{1}{1 - \ell_v(B + iu\rho_j)}.} \tag{2.2.23}$$

2.2.3 Solving the Riccati system and the final characteristic function

Closed-form solution for $B(\tau; u)$. The Riccati equation for $B(\tau; u)$,

$$B_\tau = \frac{1}{2}\sigma_v^2 B^2 + \big(\rho\sigma_v iu - \kappa\big)B - \frac{1}{2}\big(u^2 + iu\big), \qquad B(0; u) = 0,$$

is identical to the diffusion (Heston) Riccati equation, because jumps enter only the A-equation in (2.2.19). Hence B admits the standard Heston closed form (Heston, 1993). Define, for each complex $u \in \mathbb{C}$,

$$d(u) := \sqrt{\big(\kappa - \rho\sigma_v iu\big)^2 + \sigma_v^2\big(u^2 + iu\big)}, \tag{2.2.24}$$

$$g(u) := \frac{\kappa - \rho\sigma_v iu - d(u)}{\kappa - \rho\sigma_v iu + d(u)}, \tag{2.2.25}$$

where the square-root branch is chosen such that $\Re(d(u)) > 0$. This branch choice is important for numerical stability: it ensures $e^{-d(u)\tau}$ decays as τ increases and prevents spurious explosions from an incorrect branch selection.

With (2.2.24)–(2.2.25), the closed-form solution is

$$B(\tau; u) = \frac{\kappa - \rho\sigma_v iu - d(u)}{\sigma_v^2} \frac{1 - e^{-d(u)\tau}}{1 - g(u)e^{-d(u)\tau}}. \tag{2.2.26}$$

. Given $B(\tau; u)$, the function $A(\tau; u)$ solves

$$A_\tau(\tau; u) = \kappa\theta\, B(\tau; u) + \lambda\Big(M(u, B(\tau; u)) - 1 - iu\kappa_F\Big), \qquad A(0; u) = 0.$$

It is convenient to decompose A into its diffusion and jump contributions,

$$A(\tau; u) = A_{\text{diff}}(\tau; u) + A_{\text{jump}}(\tau; u), \tag{2.2.27}$$

where A_{diff} collects the terms that remain when $\lambda = 0$.

Closed-form diffusion component $A_{\text{diff}}(\tau; u)$. Using (2.2.26), the diffusion component has the standard closed form (Heston, 1993)

$$A_{\text{diff}}(\tau; u) = \frac{\kappa\theta}{\sigma_v^2}\left[\big(\kappa - \rho\sigma_v iu - d(u)\big)\tau - 2\log\left(\frac{1 - g(u)e^{-d(u)\tau}}{1 - g(u)}\right)\right]. \tag{2.2.28}$$

.

Jump component $A_{\text{jump}}(\tau; u)$. The jump contribution in (2.2.19) depends on τ only through $B(\tau; u)$. Since $A(0; u) = 0$, integrating the A-ODE over $[0, \tau]$ yields directly

$$A_{\text{jump}}(\tau; u) = \lambda \int_0^\tau \big(M\big(u, B(s; u)\big) \;-\; 1 \;-\; iu\,\kappa_F\big)\, ds, \tag{2.2.29}$$

where

$$M(u,B) := \mathbb{E}\left[e^{iuZ^y + BZ^v}\right], \qquad \kappa_F := \mathbb{E}\left[e^{Z^y} - 1\right].$$

In general, because $M(u, B(s;u))$ is a nonlinear function of $B(s;u)$, this integral does not simplify to an elementary closed form and is computed numerically.

Summarizing:

Characteristic function (SVCJ under the futures numeraire)

Let $\tau := T - t$ and $X_t = \log F_t$. The conditional characteristic function is exponential-affine:

$$\phi(u;t,T) := \mathbb{E}\left[e^{iuX_T} \mid X_t, V_t\right] = \exp\big(A(\tau;u) + B(\tau;u)V_t + iuX_t\big). \tag{2.2.30}$$

Define

$$d(u) := \sqrt{\big(\kappa - \rho\sigma_v iu\big)^2 + \sigma_v^2\big(u^2 + iu\big)},$$
$$g(u) := \frac{\kappa - \rho\sigma_v iu - d(u)}{\kappa - \rho\sigma_v iu + d(u)}, \qquad \Re(d(u)) > 0.$$

Then

$$B(\tau;u) = \frac{\kappa - \rho\sigma_v iu - d(u)}{\sigma_v^2}\,\frac{1 - e^{-d(u)\tau}}{1 - g(u)e^{-d(u)\tau}}. \tag{2.2.31}$$

$A(\tau;u) = A_{\text{diff}}(\tau;u) + A_{\text{jump}}(\tau;u)$, with

$$A_{\text{diff}}(\tau;u) = \frac{\kappa\theta}{\sigma_v^2}\left[\big(\kappa - \rho\sigma_v iu - d(u)\big)\tau - 2\log\left(\frac{1 - g(u)e^{-d(u)\tau}}{1 - g(u)}\right)\right], \tag{2.2.32}$$

$$A_{\text{jump}}(\tau;u) = \lambda \int_0^\tau \Big(M\big(u, B(s;u)\big) - 1 - iu\kappa_F\Big)\,\mathrm{d}s, \tag{2.2.33}$$

where,

$$M(u,B) := \mathbb{E}\left[e^{iuZ^y + BZ^v}\right] = \exp\left(iu\ell_y - \frac{1}{2}u^2\sigma_y^2\right)\frac{1}{1 - \ell_v\big(B + iu\rho_j\big)}, \tag{2.2.34}$$

$$\kappa_F := \mathbb{E}\left[e^{Z^y} - 1\right] = \exp\left(\ell_y + \frac{1}{2}\sigma_y^2\right)\frac{1}{1 - \ell_v\rho_j} - 1. \tag{2.2.35}$$

For completeness, the characteristic functions for the Black and Heston models under the futures price numeraire are derived in Appendix B.

2.3 Pricing inverse options under the futures numeraire

2.3.1 Inverse option payoffs and change of numeraire

Let $K > 0$ be the strike in USD per coin. The standard ("regular") USD-settled European payoffs are

$$u_{\text{call}}(F_T) = (F_T - K)^+, \qquad u_{\text{put}}(F_T) = (K - F_T)^+. \tag{2.3.1}$$

Inverse options on Deribit are quoted and settled in units of the underlying coin; they correspond to dividing the USD payoff by the terminal underlying price. Thus the inverse payoffs are (Lucic and Sepp, 2024)

$$r_{\text{call}}(F_T) = \frac{(F_T - K)^+}{F_T}, \qquad r_{\text{put}}(F_T) = \frac{(K - F_T)^+}{F_T}. \tag{2.3.2}$$

The key observation (used throughout Lucic and Sepp (2024)) is that inverse option valuation is most naturally expressed under a numeraire equal to the underlying futures price. Let F_t be the numeraire and let $\mathbb{Q}^F$ be the corresponding martingale measure. Then for any USD payoff $u(F_T)$, the time-t USD price satisfies the standard change-of-numeraire identity (Geman et al., 1995; Lucic and Sepp, 2024)

$$V_t^{\text{USD}} = F_t \, \mathbb{E}^{\mathbb{Q}^F}\left[\frac{u(F_T)}{F_T} \,\middle|\, \mathcal{F}_t \right]. \tag{2.3.3}$$

Therefore the *coin*-denominated price of the inverse payoff $r(F_T) = u(F_T)/F_T$ is simply

$$\boxed{V_t^{\text{coin}} = \mathbb{E}^{\mathbb{Q}^F}[r(F_T) \mid \mathcal{F}_t] = \frac{V_t^{\text{USD}}}{F_t}.} \tag{2.3.4}$$

In words: to price an inverse option in coin units, we can price the corresponding regular USD option and divide by the current underlying futures price.

2.3.2 Fourier pricing of the regular futures call option

We price the regular USD option using the characteristic function derived in 2.2.3 and a Fourier method. A computationally convenient approach is the FFT-based method of Carr and Madan (1999). Let $\tau = T - t$ be the time to maturity and $k = \log K$ be the log-strike. Let $C(K)$ denote the regular call price in USD. Under some regularity conditions, Carr and Madan (1999) show that

$$C(K) = \frac{e^{-\alpha k}}{\pi} \int_0^\infty \mathrm{Re}\big(e^{-iuk}\, \Psi(u)\big)\, \mathrm{d}u, \tag{2.3.5}$$

where

$$\Psi(u) = \frac{\phi(u-(\alpha+1)i;t,T)}{\alpha^2+\alpha-u^2+i(2\alpha+1)u}, \tag{2.3.6}$$

and $\phi(\cdot;t,T)$ is the characteristic function. In practice, the call price can be evaluated using FFT on a grid of strikes (Carr and Madan, 1999).

The Carr–Madan method prices options by transforming the call price as a function of log-strike $k = \log K$ into the frequency domain, where it can be expressed directly in terms of the characteristic function of $X_T = \log F_T$. Because $C(e^k)$ typically does not decay fast enough for a Fourier transform to exist, and exponentially damped call

$$\widetilde{C}(k) := e^{\alpha k} C(e^k), \qquad \alpha > 0,$$

is introduced, with α chosen so that $\widetilde{C} \in L^2(\mathbb{R})$.

We then take the Fourier transform of $\widetilde{C}$ (with respect to k),

$$\widehat{\widetilde{C}}(u) := \int_{-\infty}^{\infty} e^{iuk}\, \widetilde{C}(k)\, \mathrm{d}k, \qquad u \in \mathbb{R}. \tag{2.3.7}$$

Starting from $C(e^k) = \mathbb{E}[(F_T - e^k)^+ \mid \mathcal{F}_t]$ and substituting into (2.3.7), one obtains an expression of the form

$$\widehat{\widetilde{C}}(u) = \mathbb{E}\left[\int_{-\infty}^{\infty} e^{iuk} e^{\alpha k} (F_T - e^k)^+\, \mathrm{d}k \,\middle|\, \mathcal{F}_t\right].$$

Under the same integrability conditions that justify the damping, the order of expectation and integration can be exchanged. The resulting inner integral can be evaluated in closed form and reduces the transform $\widehat{\widetilde{C}}(u)$ to a rational factor times an exponential moment of X_T, which is precisely the characteristic function evaluated at a complex argument. This yields the closed-form representation

$$\widehat{\widetilde{C}}(u) = \Psi(u), \tag{2.3.8}$$

where $\Psi(u)$ is given in (2.3.6).

Finally, the damped call is recovered by the inverse Fourier transform,

$$\widetilde{C}(k) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-iuk}\, \widehat{\widetilde{C}}(u)\, \mathrm{d}u = \frac{1}{\pi}\int_{0}^{\infty} \mathrm{Re}\big(e^{-iuk}\Psi(u)\big)\, \mathrm{d}u, \tag{2.3.9}$$

and undoing the damping gives the desired call price

$$C(K) = C(e^k) = e^{-\alpha k}\widetilde{C}(k),$$

which is (2.3.5).

In practice, the integral in (2.3.5) is approximated on an evenly spaced frequency grid $u_j = j\Delta u$, $j = 0, \ldots, N-1$. Using the trapezoidal rule, the inverse transform becomes a discrete Fourier transform mapping the sampled values $\Psi(u_j)$ to call prices on an evenly spaced log-strike grid $k_m = k_0 + m\Delta k$, $m = 0, \ldots, N-1$, with the grid spacings linked by $\Delta k\, \Delta u = 2\pi/N$. This is precisely the structure computed efficiently by the FFT, allowing simultaneous pricing of a large set of strikes in $O(N \log N)$ time.

For a more detailed discussion of the Carr-Madan procedure, see Appendix A. The actual implementation of the Carr-Madan FFT pricer is available on GitHub.

2.3.3 Inverse call and put option price

Given $C(K)$ in USD units, the inverse call option price in coin unit is

$$C^{\text{coin}}(K) = \frac{C(K)}{F_t}, \tag{2.3.10}$$

which is the direct application of (2.3.4).

The put inverse option on the futures can be calculated using the following put–call parity for inverse options[2]:

Theorem (Put–call parity for inverse options)**.** *Let F_t be the time-t price of a futures contract maturing at T, and let $C_t(K,T)$ and $P_t(K,T)$ denote the time-t prices of European call and put options with strike K and maturity T written on this futures. Define the futures-numeraire prices*

$$\widehat{C}_t(K,T) := \frac{C_t(K,T)}{F_t}, \qquad \widehat{P}_t(K,T) := \frac{P_t(K,T)}{F_t}.$$

Then, under the pricing measure associated with the futures numeraire, the following put–call parity holds:

$$\boxed{\widehat{C}_t(K,T) - \widehat{P}_t(K,T) = 1 - \frac{K}{F_t}.} \tag{2.3.11}$$

For the proof of the above theorem, see Appendix C.

2.4 Model deltas and net deltas

2.4.1 Regular delta under the pricing model

We first define the usual delta of the corresponding regular option with respect to the dated futures price:

$$\Delta_t^{\text{reg}} := \frac{\partial V_t^{\text{USD}}}{\partial F_t}. \tag{2.4.1}$$

[2] Assuming zero discounting, following (Lucic and Sepp, 2024)

For a call option priced under the futures numeraire, affine pricing formulas admit the standard representation

$$C_t = F_t \Pi_{1,t} - K \Pi_{2,t}, \tag{2.4.2}$$

where

$$\Pi_{1,t} = Q^F(F_T > K \mid \mathcal{F}_t)$$

is the probability of finishing in the money under the futures-price measure Q^F, and

$$\Pi_{2,t} = F_t \, \mathbb{E}^{Q^F}\left[\frac{\mathbf{1}_{\{F_T > K\}}}{F_T} \,\middle|\, \mathcal{F}_t \right].$$

If one additionally introduces the cash-numeraire pricing measure, then $\Pi_{2,t}$ coincides with the corresponding risk-neutral in-the-money probability. Under standard regularity conditions this implies

$$\Delta^{\text{reg}}_{t,\text{call}} = \Pi_{1,t}. \tag{2.4.3}$$

For puts, the regular delta can either be computed directly from the put pricing formula or recovered from regular put–call parity. Under zero discounting,

$$C_t - P_t = F_t - K, \tag{2.4.4}$$

and differentiation with respect to F_t yields

$$\Delta^{\text{reg}}_{t,\text{put}} = \Delta^{\text{reg}}_{t,\text{call}} - 1. \tag{2.4.5}$$

In the empirical implementation below, the model-specific object that is computed first is therefore always the regular delta Δ^{reg}_t, which is then converted into the premium-adjusted net delta.

2.4.2 Inverse delta as the sensitivity of the coin price

For inverse options, the object actually observed and hedged in practice is the coin-denominated price

$$V^{\text{coin}}_t = \frac{V^{\text{USD}}_t}{F_t}. \tag{2.4.6}$$

Differentiating with respect to F_t gives the *inverse* delta

$$\frac{\partial V^{\text{coin}}_t}{\partial F_t} = \frac{1}{F_t}\frac{\partial V^{\text{USD}}_t}{\partial F_t} - \frac{V^{\text{USD}}_t}{F_t^2} = \frac{1}{F_t}\left(\Delta^{\text{reg}}_t - \frac{V^{\text{USD}}_t}{F_t}\right). \tag{2.4.7}$$

Using $V^{\text{coin}}_t = V^{\text{USD}}_t / F_t$, this becomes

$$\frac{\partial V^{\text{coin}}_t}{\partial F_t} = \frac{1}{F_t}\left(\Delta^{\text{reg}}_t - V^{\text{coin}}_t\right). \tag{2.4.8}$$

Specializing to calls and puts,

$$\frac{\partial C_t^{\text{coin}}}{\partial F_t} = \frac{1}{F_t}\left(\Delta_{t,\text{call}}^{\text{reg}} - C_t^{\text{coin}}\right), \tag{2.4.9}$$

$$\frac{\partial P_t^{\text{coin}}}{\partial F_t} = \frac{1}{F_t}\left(\Delta_{t,\text{put}}^{\text{reg}} - P_t^{\text{coin}}\right). \tag{2.4.10}$$

Equations (2.4.9)–(2.4.10) show why inverse options cannot be hedged using the textbook regular delta alone: the coin price depends on the futures level both through the USD option value and through the numeraire conversion by $1/F_t$.

2.4.3 Net delta

Following Lucic and Sepp (2024), the premium-adjusted or *net delta* is defined by

$$\Delta_t^{\text{net}} := \Delta_t^{\text{reg}} - V_t^{\text{coin}}. \tag{2.4.11}$$

Equivalently, for calls and puts,

$$\Delta_{t,\text{call}}^{\text{net}} = \Delta_{t,\text{call}}^{\text{reg}} - C_t^{\text{coin}}, \qquad \Delta_{t,\text{put}}^{\text{net}} = \Delta_{t,\text{put}}^{\text{reg}} - P_t^{\text{coin}}. \tag{2.4.12}$$

With this notation, the inverse delta is simply the net delta rescaled by $1/F_t$:

$$\frac{\partial V_t^{\text{coin}}}{\partial F_t} = \frac{\Delta_t^{\text{net}}}{F_t}. \tag{2.4.13}$$

This identity is the central hedging relation used throughout the thesis. It implies that for a small move in the dated futures price,

$$dV_t^{\text{coin}} \approx \frac{\Delta_t^{\text{net}}}{F_t}\, dF_t. \tag{2.4.14}$$

In words, the net delta is the first-order exposure of the inverse option value to futures moves, expressed in the natural units relevant for inverse-futures hedging.

2.4.4 From net delta to hedge notional

To connect the option sensitivity to a hedge in inverse futures or perpetual futures, consider a position with fixed USD notional $N_{\text{USD},t}$. For an inverse futures-type contract, the coin P&L over a small interval is driven by changes in $1/F_t$. Using the first-order expansion

$$\frac{1}{F_t} - \frac{1}{F_t + dF_t} \approx \frac{dF_t}{F_t^2}, \tag{2.4.15}$$

the hedge P&L in coin units is approximately

$$d\Pi_t^{\text{hedge}} \approx N_{\text{USD},t}\frac{dF_t}{F_t^2}. \tag{2.4.16}$$

Now suppose the investor holds q units of the inverse option, where $q > 0$ corresponds to a long option position and $q < 0$ to a short option position. The option's first-order coin P&L is

$$d\Pi_t^{\mathrm{opt}} \approx q\, dV_t^{\mathrm{coin}} \approx q\, \frac{\Delta_t^{\mathrm{net}}}{F_t}\, dF_t. \tag{2.4.17}$$

A first-order delta-neutral hedge sets

$$d\Pi_t^{\mathrm{opt}} + d\Pi_t^{\mathrm{hedge}} \approx 0,$$

which yields

$$q\, \frac{\Delta_t^{\mathrm{net}}}{F_t}\, dF_t + N_{\mathrm{USD},t} \frac{dF_t}{F_t^2} \approx 0.$$

Therefore the hedge notional implied by net delta is

$$N_{\mathrm{USD},t}^{\mathrm{net}} = -\, q\, F_t\, \Delta_t^{\mathrm{net}}. \tag{2.4.18}$$

Equation (2.4.18) is the basic mapping from the model-implied net delta to the USD notional of the inverse hedge. In particular, for a *short* option position ($q = -1$),

$$N_{\mathrm{USD},t}^{\mathrm{net}} = F_t\, \Delta_t^{\mathrm{net}}. \tag{2.4.19}$$

For comparison, one may also define the conventional, non-premium-adjusted hedge notional based on the regular delta alone:

$$N_{\mathrm{USD},t}^{\mathrm{reg}} = -\, q\, F_t\, \Delta_t^{\mathrm{reg}}. \tag{2.4.20}$$

This gives two distinct hedge conventions:

- **regular-delta hedge:** uses Δ_t^{reg} and ignores the inverse premium adjustment;
- **net-delta hedge:** uses Δ_t^{net} and is the theoretically correct first-order hedge ratio for inverse options.

The remainder of the chapter and the empirical analysis in Chapter 4 use this notation. Section 2.5 translates (2.4.18) into a practical hedge with the more liquid inverse perpetual futures contract.

2.5 Hedging inverse options with perpetual futures

Section 2.4 established that the economically relevant first-order hedge ratio for an inverse option is the net delta, and that the corresponding inverse-futures hedge notional is obtained by multiplying the net delta by the dated futures price. We now translate this result into the actual hedging instrument used in practice: the perpetual futures contract.

2.5.1 Why hedge with the perpetual rather than the dated future?

On Deribit, the listed option is written on the dated futures contract $F_t^{(T)}$ with the same expiry as the option. In principle, the cleanest hedge would therefore use the same dated futures contract. In practice, however, perpetual futures are typically much more liquid than individual dated futures, especially at shorter horizons. Also, for inverse options with synthetic futures underlyings, the underlying may not be tradeable. For these reasons, dynamic hedging is usually implemented with the perpetual rather than the true underlying dated futures contract.

Let $F_t^{\perp}$ denote the price of the inverse perpetual futures contract, quoted in USD per coin. The dated futures price $F_t^{(T)}$ and the perpetual price $F_t^{\perp}$ are generally close, but not identical. Consequently, hedging an option written on $F_t^{(T)}$ with a position in $F_t^{\perp}$ introduces *basis risk*. This basis risk reflects differences between the dated futures and the perpetual that arise from carry, funding, liquidity, and microstructure effects.

The theoretical delta calculations in Section 2.4 are therefore computed with respect to the option underlying $F_t^{(T)}$, while the actual hedge P&L is realized through the perpetual $F_t^{\perp}$.

2.5.2 Discrete-time P&L of the inverse perpetual

Consider an inverse perpetual position with fixed USD notional N_{USD,t_i} held over the interval $[t_i, t_{i+1}]$. Its coin-denominated P&L is

$$\Pi^{\perp}(t_i, t_{i+1}; N_{\mathrm{USD},t_i}) = N_{\mathrm{USD},t_i} \left(\frac{1}{F_{t_i}^{\perp}} - \frac{1}{F_{t_{i+1}}^{\perp}} \right). \tag{2.5.1}$$

This is the discrete-time building block for all hedging analysis in Chapter 4.

Now consider an inverse option position of size q. From Section 2.4, the hedge notional implied by the theoretically correct premium-adjusted hedge ratio is

$$N_{\mathrm{USD},t_i}^{\mathrm{net}} = -\,q\,F_{t_i}^{(T)}\,\Delta_{t_i}^{\mathrm{net}}. \tag{2.5.2}$$

Likewise, the benchmark hedge based on the conventional regular delta is

$$N_{\mathrm{USD},t_i}^{\mathrm{reg}} = -\,q\,F_{t_i}^{(T)}\,\Delta_{t_i}^{\mathrm{reg}}. \tag{2.5.3}$$

For the case of a short option position ($q = -1$), these reduce to

$$N_{\mathrm{USD},t_i}^{\mathrm{net}} = F_{t_i}^{(T)}\,\Delta_{t_i}^{\mathrm{net}}, \qquad N_{\mathrm{USD},t_i}^{\mathrm{reg}} = F_{t_i}^{(T)}\,\Delta_{t_i}^{\mathrm{reg}}. \tag{2.5.4}$$

3. Data and Calibration Results

This chapter describes the empirical dataset used in the study, the preprocessing and liquidity filters applied to Deribit option quotes, the calibration methodology employed to fit the Black, Heston, and SVCJ models to observed inverse option prices, and the calibration results.

The headline empirical result is that the SVCJ model gives consistent improvements in out-of-sample pricing accuracy over the Black and Heston baselines.

3.1 Dataset description

3.1.1 Data source and collection procedure

All market data are sourced from the Deribit public API[1]. The dataset is constructed from timestamped snapshots of the BTC and ETH option chains, augmented with snapshots of perpetual futures quotes. The data collection period is from 30-12-2025 till 08-04-2026. Option surface snapshots are collected roughly 4 times daily at 6 hour intervals, giving a total of 388 snapshots. Each snapshot time is denoted by t, and the corresponding panel of options is used for cross-sectional calibration at that time.[2]

3.1.2 Option snapshot fields

Each option snapshot is stored as a CSV file[3], with one row per listed option instrument. Table 3.1.1 lists the fields stored for each option.

Table 3.1.1: Fields stored in the option snapshot dataset.

Field	**Description**
`timestamp`	Snapshot timestamp (UTC), representing the observation time t.
`currency`	Underlying currency (BTC or ETH).
`instrument_name`	Deribit instrument identifier (unique option name).

[1] Deribit API documentation
[2] The code used for collecting the data is available on GitHub
[3] Available on GitHub

Field	Description
option_type	Option type: call or put.
strike	Strike K in USD per coin.
expiry_datetime	Expiration datetime (UTC).
time_to_maturity	Time to maturity T in years, computed at collection time.
bid_price, ask_price	Best bid/ask option premia (coin-denominated).
mid_price	Mid premium (coin-denominated). If unavailable, computed as $(\texttt{bid} + \texttt{ask})/2$.
mark_price	Deribit mark price (coin-denominated).
bid_ask_spread	ask_price − bid_price (coin-denominated).
implied_volatility	Quoted mark implied volatility (decimal).
delta, vega	Option Greeks.
open_interest	Open interest (contract units as reported by Deribit).
volume, volume_usd	Trading volume as reported by Deribit (coin units and USD).
index_price	Spot index level (USD per coin).
underlying	Underlying level (typically a futures/synthetic forward).
underlying_name	Underlying identifier associated with underlying. If equal to index_price, the underlying is the spot index; otherwise it typically refers to a dated future or a synthetic forward.
futures_price	Forward/futures proxy used in calibration. This field is populated when underlying_name is not index_price; otherwise it may be missing.
futures_instrument_name	Instrument name associated with futures_price (when available).

3.1.3 Perpetual futures snapshot fields

In addition to the option surface, a separate CSV file[4] stores a snapshot of perpetual futures quotes, primarily intended for subsequent hedging and performance evaluation. Table 3.1.2 lists the stored fields.

[4]Also available on GitHub

Table 3.1.2: Fields stored in the perpetual futures snapshot dataset.

Field	Description
obs_datetime	Observation time (UTC).
coin	Underlying coin: BTC or ETH.
perp_futures_mark_price	Perpetual futures mark price (USD per coin).

3.1.4 Pre-calibration cleaning and liquidity filters

Raw snapshots contain stale quotes, wide markets, and instruments with negligible trading activity. Since calibration is performed by matching model prices to observed mid quotes, the dataset is filtered to retain reasonably liquid instruments. The filtering procedure applied to each snapshot consists of the following steps:

1. **Basic validation.** Ensure all required fields are present (currency, option type, strike, maturity, bid/ask quotes, underlying price, vega, open interest, and expiry identifier).

2. **Maturity filtering.** Retain only options with strictly positive time-to-maturity and enforce a minimum maturity threshold:

$$T \geq \frac{1}{365} \text{ years.}$$

3. **Quote quality.** Require strictly positive bid and ask quotes, and enforce $\texttt{ask\_price} \geq \texttt{bid\_price}$.

4. **Activity thresholds.** Enforce minimum levels of open interest and vega. The default values are specified below:

$$\texttt{open\_interest} \geq 1, \qquad \texttt{vega} > 0.$$

5. **Relative spread filter.** Define the clean mid price and relative spread as

$$P^{mid} = \tfrac{1}{2}(\texttt{bid\_price} + \texttt{ask\_price}), \qquad \texttt{rel\_spread} = \frac{\texttt{ask\_price} - \texttt{bid\_price}}{P^{mid}},$$

   and impose the cutoff $\texttt{rel\_spread} \leq 0.5$.

6. **Forward proxy and moneyness.** For each expiry bucket, define a forward proxy F_0 as the median futures_price observed in that bucket (the observed futures price may vary slightly within each expiry bucket due to lags in obtaining the data), and compute moneyness as K/F_0. By default, keep options with

$$0.5 \leq \frac{K}{F_0} \leq 2.0.$$

7. **Instrument type.** Drop if not `call` or `put`.

To quantify the effect of the cleaning procedure, Table 3.1.3 compares the number of raw and retained quotes before and after filtration. Out of 614,870 raw option quotes in the pooled sample, 355,552 remain after filtering, implying an overall retention rate of 57.8%. Retention is somewhat higher for BTC than for ETH: 60.3% of BTC quotes are retained, against 55.2% for ETH. The same pattern appears at the snapshot level. The median BTC panel falls from 854 raw quotes to 480.5 filtered quotes, while the median ETH panel falls from 772 to 427. In pooled terms, the median snapshot shrinks from 1,622 to 883 quotes. This is a substantial reduction, but it is economically reasonable: the objective is not to preserve the largest possible sample, but to retain quotes that are sufficiently liquid and internally consistent for cross-sectional calibration. Figure 3.1.1 visualizes this information dynamically.

Table 3.1.3: Raw and filtered quote counts by currency.

Currency	Snapshots	Raw quotes	Filtered quotes	Removed quotes	Retention rate (%)	Median raw / snapshot	Median filtered / snapshot
BTC	388	316,560	191,033	125,527	60.35	854.0	480.5
ETH	388	298,310	164,519	133,791	55.15	772.0	427.0
Pooled	388	614,870	355,552	259,318	57.83	1,622.0	883.0

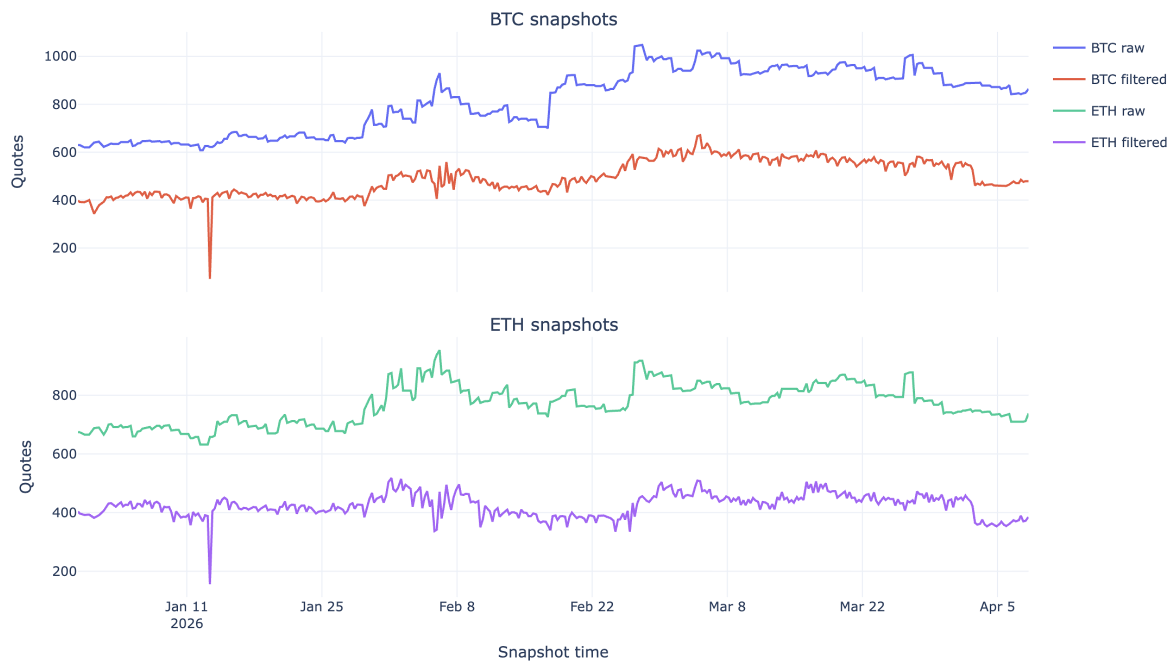

Figure 3.1.1: Raw versus filtered quote counts over time.

Table 3.1.4 shows where this attrition occurs. The largest single reduction comes from the open-interest screen, which removes 98,549 pooled quotes, or 17.9% of the sample surviving the previous step. The moneyness restriction $0.5 \leq K/F_0 \leq 2.0$ is the second most restrictive filter, removing a further 62,908 quotes, or 15.0% of the sample remaining at that stage. The relative-spread filter also matters, excluding 29,701 pooled quotes. By contrast, the forward-proxy requirement removes very few observations, and the checks $ask \geq bid$, $vega \geq 0$, and the call/put type restriction are essentially non-binding once the earlier filters have been imposed. Economically, this implies that most of the sample reduction is driven by liquidity and extreme-moneyness exclusions rather than by basic data-quality failures.

Table 3.1.4: Main sources of sample attrition.

Stage	BTC removed	ETH removed	Pooled removed	Pooled removal share of previous (%)
$T \geq 1$ day	18,424	17,798	36,222	5.89
$bid, ask > 0$	13,878	15,646	29,524	5.10
$open\ interest \geq 1$	54,164	44,385	98,549	17.95
$rel.\ spread \leq 0.5$	14,901	14,800	29,701	6.59
Forward proxy available	1,222	1,192	2,414	0.57
$0.5 \leq K/F_0 \leq 2.0$	22,938	39,970	62,908	15.03

3.1.5 EDA of the filtered calibration data

Before turning to model calibration, it is useful to describe the structure of the filtered calibration sample. Table 3.1.5 summarizes the sample by currency. BTC panels are somewhat denser than ETH panels: as stated previously, the median BTC snapshot contains 480.5 usable contracts, against 427.0 for ETH. The filtered sample is close to balanced between calls and puts for both assets, and the median moneyness in both markets is slightly above one, indicating that the retained contracts cluster around the underlying futures.

Table 3.1.5: Filtered pooled calibration sample by currency.

Currency	Snapshots	Quotes	Calls	Puts	Median options per snapshot	Median maturity (days)	Median K/F_0	Median relative spread
BTC	388	191033	95682	95351	480.5	47.4	1.042	0.033
ETH	388	164519	83738	80781	427.0	42.4	1.053	0.036

Figure 3.1.2 shows the maturity and moneyness distributions for the pooled filtered sam-

ple. Two features matter for model design: (i) the option universe is heavily tilted toward short maturities, which is where skew and jump effects typically become more apparent, but still exhibits a meaningful right tail extending to roughly one year; and (ii) the moneyness distribution is centred near the futures, yet remains wide enough that any successful model must fit both the centre of the smile and the wings. This is precisely where the flexibility of stochastic volatility and jumps should matter most.

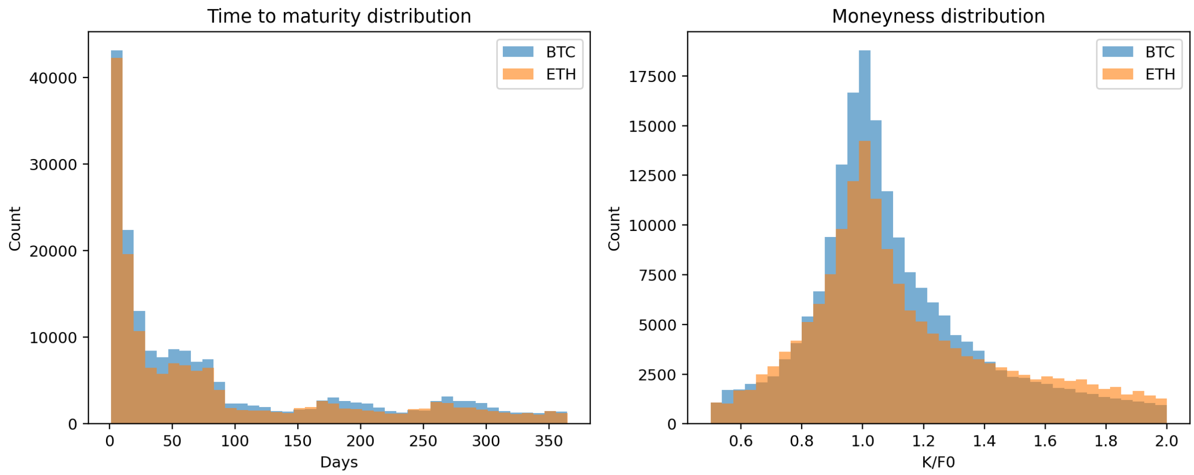

Figure 3.1.2: Distribution of filtered option quotes by maturity and moneyness.

3.2 Calibration methodology

Model parameters are estimated via repeated cross-sectional calibration. At each snapshot time t, let $\{(K_i, T_i, F_{0,i}, P_i^{mkt})\}_{i=1}^n$ denote the filtered set of option quotes, where K_i is the strike, T_i the time to maturity, $F_{0,i}$ the futures proxy, and P_i^{mkt} the observed inverse option mid-price in coin units. For a given model with parameter vector Θ, the pricing engine produces model-implied inverse option prices $P_i^{model}(\Theta)$ in the same units. The calibration objective is to choose $\widehat{\Theta}$ to minimize a weighted least-squares discrepancy between P^{model} and P^{mkt}.

The calibration is carried out separately for the Black, Heston, and SVCJ models. All models are priced under the futures price numeraire consistent with the Deribit inverse-quote convention, and calibration is performed in coin units to match observed market premia.

Because calibration is repeated independently on a dense time series of snapshots, the raw cross-sectional objective may produce economically implausible parameter jumps from one snapshot to the next, especially when the model has many parameters that lead to similar prices. To reduce this effect, the implementation incorporates *temporal smoothing*: when calibrating a snapshot at time t, the optimizer is encouraged to remain close to the parameter vector obtained at the previous snapshot via an L_2-type penalty.

3.2.1 Objective function, time smoothing and weighting

Let $w_i \geq 0$ denote a weight assigned to quote i. The residual for quote i is defined as

$$r_i(\Theta_t) \;=\; w_i\big(P_i^{model}(\Theta_t) - P_i^{mkt}\big),$$

and the estimator solves the weighted least-squares problem

$$\widehat{\Theta}_t \;\in\; \arg\min_{\Theta_t} \left\{ \sum_{i=1}^{n} r_i(\Theta_t)^2 + \lambda_{\mathrm{TS}} \big\| \Theta_t - \widehat{\Theta}_{t-1} \big\|_2^2 \right\}, \tag{3.2.1}$$

where the additional term $\lambda_{\mathrm{TS}}\big\|\Theta_t - \widehat{\Theta}_{t-1}\big\|_2^2$ enforces temporal smoothing over the previous snapshot's calibrated parameters.

Economically, temporal smoothing should be interpreted as a stabilization device for the sequence of snapshot-level estimates, rather than as a claim that the latent model parameters follow a specified dynamic process. In the empirical analysis below, several values of λ_{TS} are compared in order to assess the trade-off between cross-sectional fit and temporal stability of the calibrated parameter paths.

Weights are constructed multiplicatively from three components: a spread-based term, a vega-based term, and an open-interest term. Let $\texttt{spread}_i = \texttt{ask}_i - \texttt{bid}_i$ denote the bid–ask spread in coin units, and let $\texttt{vega}_i$ and $\texttt{oi}_i$ denote the vega and open interest reported by Deribit. The weight used takes the form

$$w_i \;=\; \min\Big\{ \big(\texttt{spread}_i + \varepsilon_s\big)^{-p_s} \big(\texttt{vega}_i + \varepsilon_o\big)^{p_v} \big(\texttt{oi}_i + \varepsilon_o\big)^{p_{oi}},\; w_{\max} \Big\}, \tag{3.2.2}$$

where the constants are set by default to

$$p_s = 1, \quad p_v = 0, \quad p_{oi} = 0, \quad \varepsilon_s = 10^{-6}, \quad \varepsilon_o = 10^{-12}, \quad w_{\max} = 10^6.$$

The spread component downweights wide markets, while vega and open interest can be interpreted as proxies for the informational content and liquidity of the quote.

In the baseline specification, only the spread component is included in the weight (as $p_v = p_{oi} = 0$). This is done so that the model is exposed to relatively illiquid and far-from-the-money options during calibration, as this is the setting where the stochastic volatility and jump components are most useful in improving pricing accuracy over the baseline Black and Heston models. The vega and open interest components are left for additional robustness checks in further analysis.

3.2.2 Model parameters and bounds

This section lists the parameters for each model, provides brief economic interpretations, and reports the admissible bounds enforced during calibration.

Black model

The Black model has a single parameter σ, the constant volatility of log-futures returns under the forward measure. The calibration bounds are:

$$\sigma \in [10^{-4},\, 5].$$

Heston model

The Heston stochastic volatility model is parameterized by

$$\Theta_{\mathrm{H}} = (\kappa, \theta, \sigma_v, \rho, v_0),$$

where κ is the variance mean reversion speed, θ is the long-run variance level, σ_v is the volatility of variance ("vol-of-vol"), ρ is the instantaneous correlation between the return and variance shocks, and v_0 is the initial variance.

The calibration bounds are:

$$\kappa \in [10^{-4},\, 50], \qquad \theta \in [10^{-6},\, 5], \qquad \sigma_v \in [10^{-4},\, 10], \qquad v_0 \in [10^{-6},\, 5],$$

and

$$\rho \in [\tanh(-5),\, \tanh(5)] \approx [-0.99991,\, 0.99991].$$

SVCJ model

The SVCJ model extends Heston by adding correlated jumps in returns and variance. The parameter vector is

$$\Theta_{\mathrm{SVCJ}} = (\kappa, \theta, \sigma_v, \rho, v_0, \lambda, \ell_y, \sigma_y, \ell_v, \rho_j),$$

where $(\kappa, \theta, \sigma_v, \rho, v_0)$ are as in Heston, λ is the jump intensity, (ℓ_y, σ_y) control the return-jump distribution, ℓ_v is the (positive) mean parameter of the variance jump distribution, and ρ_j is a dependence parameter governing return/variance jump correlation.

The calibration bounds are:

$$\kappa \in [10^{-4},\, 50], \quad \theta \in [10^{-6},\, 5], \quad \sigma_v \in [10^{-4},\, 10], \quad v_0 \in [10^{-6},\, 5],$$

$$\lambda \in [10^{-6}, 10], \quad \ell_y \in [-5, 5], \quad \sigma_y \in [10^{-4}, 5], \quad \ell_v \in [10^{-6}, 10],$$

and

$$\rho \in [\tanh(-5), \tanh(5)], \qquad \rho_j \in [\tanh(-5), \tanh(5)].$$

3.2.3 Additional constraints

In addition to simple parameter bounds, the calibration incorporates two economically and numerically important admissibility conditions.

Feller-type condition (Heston and SVCJ)

For both Heston and SVCJ, the Feller condition (Heston, 1993) ensures strict positivity of the variance process and improves numerical stability of Fourier-based pricing. The condition is

$$\sigma_v^2 \;\le\; 2\kappa\theta, \tag{3.2.3}$$

and violations are discouraged via a soft penalty. Concretely, define the violation magnitude

$$\Delta_F(\Theta_t) \;=\; \max\left\{0,\; \sigma_v^2 - 2\kappa\theta\right\}.$$

The penalty contribution is appended to the residual vector with scaling constant c_F:

$$r_F(\Theta_t) \;=\; c_F\, \Delta_F(\Theta_t), \qquad c_F = 100.$$

Moment/stability condition (SVCJ)

For SVCJ, characteristic-function evaluation imposes additional restrictions to ensure the jump component remains well-defined. The implementation penalizes violations of

$$1 - \ell_v \rho_j \;\ge\; \varepsilon_J, \tag{3.2.4}$$

where $\varepsilon_J = 10^{-6}$ in the baseline. Define

$$\Delta_J(\Theta_t) \;=\; \max\left\{0,\; \varepsilon_J - \big(1 - \ell_v \rho_j\big)\right\},$$

and append the penalty residual

$$r_J(\Theta_t) \;=\; c_J\, \Delta_J(\Theta_t), \qquad c_J = 100.$$

3.2.4 Robust handling of numerical failures

Fourier-based pricing may occasionally return non-finite values for extreme parameter choices (e.g. due to overflow in exponentials). To prevent the optimizer from stalling, any pricing failure or non-finite model price is mapped to a fixed penalty residual. Specifically, for a quote i with non-finite $P_i^{model}(\Theta_t)$, the residual is set to

$$r_i(\Theta_t) \;=\; w_i \cdot \texttt{penalty\_coin}, \qquad \texttt{penalty\_coin} = 10,$$

and the optimization proceeds.

3.2.5 Calibration implementation outline

Calibration is implemented as a bounded nonlinear least-squares problem, using the `scipy.optimize.least_squares` function's Trust Region Reflective (TRF) algorithm for solving nonlinear least-squares (Branch et al., 1999) from the SciPy package for Python (Virtanen et al., 2020). Efficiency is achieved by exploiting the FFT structure: for a given maturity, a single Carr–Madan FFT call produces a grid of call prices across strikes, after which put prices are obtained via inverse put–call parity. The calibration routine therefore groups options by expiry and prices each group with one FFT evaluation per function call.[5]

In addition, the FFT log-strike grid is centered dynamically per expiry bucket. Let N and η denote the FFT grid size and frequency spacing. The log-strike spacing is $\lambda = 2\pi/(N\eta)$, and the minimum log-strike parameter b is chosen as

$$b \;=\; \log\!\big(\mathrm{median}(K)\big) \;-\; \tfrac{1}{2}N\lambda,$$

so that the grid is centered around the typical strikes observed for that expiry. This improves interpolation quality while keeping FFT grids compact.

Algorithm 1 summarizes the procedure executed at each snapshot time t.

[5] The calibration script is available on GitHub

Algorithm 1 Cross-sectional calibration at snapshot time t.

1: **Input:** raw snapshot data; model family; FFT base parameters; weighting configuration.
2: Filter snapshot to liquid options (Section 3.1.4); compute clean mid prices P_i^{mkt}.
3: For each expiry bucket, compute forward proxy F_0 as median `futures_price`; compute moneyness K/F_0.
4: Build a per-expiry pricing plan: store indices, strikes, maturity T, forward proxy F_0, and FFT grid parameters (with dynamic centering).
5: Define weights w_i according to (3.2.2).
6: Define residual vector $r(\Theta_t)$ consisting of weighted squared difference between observed and model prices plus the time smoothing term $r_i(\Theta_t)$, and constraint penalties $r_F(\Theta_t)$, $r_J(\Theta_t)$ where applicable.
7: Solve the bounded least-squares problem (3.2.1).
8: **Output:** calibrated parameters $\widehat{\Theta}_t$; fit diagnostics (RMSE, MAE).

3.3 Calibration results

3.3.1 Choice of the temporal smoothing parameter λ_{TS}

Figure 3.3.1 summarizes the temporal smoothing trade-off between out-of-sample pricing error, number of function evaluations and parameter stability. The left panel shows the two direct costs of calibration at each value of λ_{TS}: pricing accuracy, measured by the mean out-of-sample test RMSE, and optimisation effort, measured by the mean number of function evaluations. The right panel shows how smooth the resulting parameter paths are over time.

The roughness index in the right panel is constructed to summarize the time-series instability of the calibrated SVCJ parameters. For each currency and each selected parameter, we first compute the median absolute change between consecutive snapshots, and then scale it by the parameter's time-series dispersion, measured by the interquartile range. The index plotted in Figure 3.3.1 is the average of these standardized median changes across the parameters of the SVCJ model and across BTC and ETH. A lower value therefore means that, relative to its own scale, a parameter moves less from one snapshot to the next, so the calibration is smoother and more stable over time.

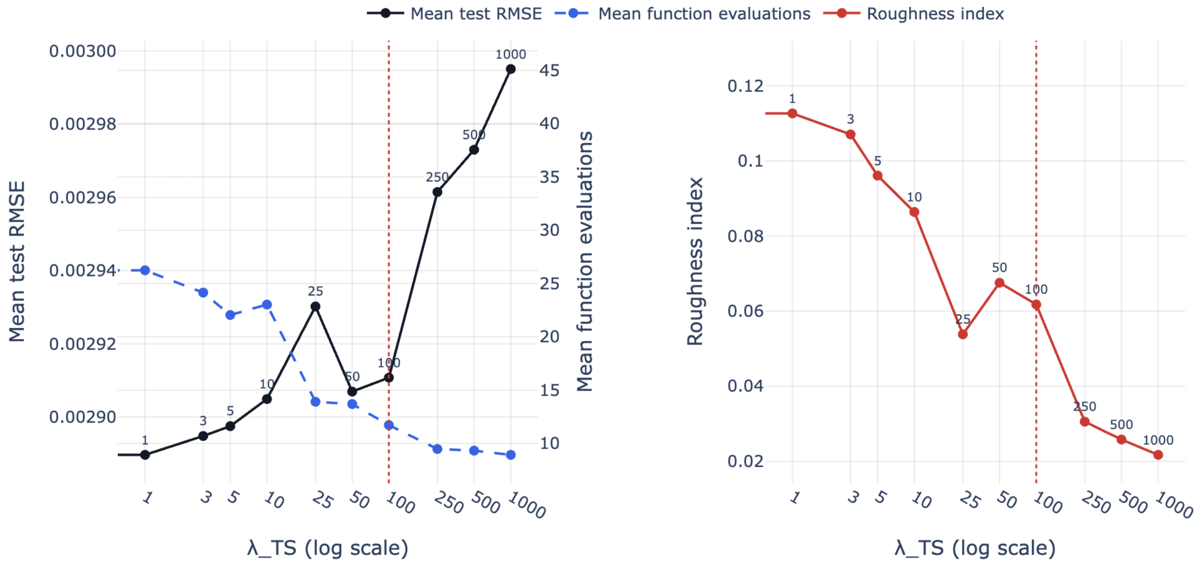

Figure 3.3.1: Regularisation trade-off across the SVCJ sweep. The left panel reports the mean out-of-sample test RMSE and the mean number of function evaluations across snapshots. The right panel reports an aggregate roughness index for the calibrated SVCJ parameter paths. The vertical dashed line marks the final baseline choice, $\lambda_{TS} = 100$.

The left panel of Figure 3.3.1 shows that the raw fit optimum is attained at the very low end of the grid. The smallest mean test RMSE is achieved at $\lambda_{TS} = 1$, where the average snapshot-level test RMSE is 0.002890. But the same point in the figure also shows why this is not an attractive baseline. The blue series indicates a mean of 26.24 function evaluations, while the right panel reports a roughness index of 0.1127. In other words, the gain in RMSE relative to the rest of the grid is small, yet it comes with both higher computational cost and materially rougher parameter paths.

The practically relevant part of the figure is the middle of the grid, especially $\lambda_{TS} \in \{25, 50, 100\}$. Figure 3.3.1 shows that $\lambda_{TS} = 25$ is a transition point rather than a natural final choice: on the one hand, the right panel shows that moving from $\lambda_{TS} = 10$ to $\lambda_{TS} = 25$ produces a sharp reduction in roughness, from 0.0864 to 0.0538; on the other hand, the left panel shows that this smoothing gain is not accompanied by superior pricing performance: the mean test RMSE at $\lambda_{TS} = 25$ is 0.002930, which is worse than at both $\lambda_{TS} = 50$ and $\lambda_{TS} = 100$.

The actual decision is therefore between $\lambda_{TS} = 50$ and $\lambda_{TS} = 100$. In the left panel, the black RMSE line is essentially flat between these two values: the mean test RMSE changes only from 0.002907 at $\lambda_{TS} = 50$ to 0.002911 at $\lambda_{TS} = 100$, which is economically negligible. But the blue optimisation-cost line continues to fall, from 13.69 to 11.71 mean function evaluations. At the same time, the right panel shows that the roughness index also improves, declining from 0.0676 to 0.0618. So relative to $\lambda_{TS} = 50$, the choice $\lambda_{TS} = 100$ preserves virtually all of the pricing accuracy while improving both computational efficiency and path smoothness.

The figure also shows why it is not sensible to push the penalty much further. For $\lambda_{TS} \geq 250$, the right panel continues to move downward, so the parameter paths do become smoother, and the blue line in the left panel also declines slightly further. But the black RMSE line then starts to drift upward significantly: the mean test RMSE rises to 0.002961 at $\lambda_{TS} = 250$, 0.002973 at $\lambda_{TS} = 500$, and 0.002995 at $\lambda_{TS} = 1000$. This is the point at which the regularisation begins to smooth too aggressively and the pricing fit materially deteriorates.

For the remainder of the analysis, we therefore fix $\lambda_{TS} = 100$ as the common baseline for temporal smoothing in the calibration problem.

3.3.2 RMSE, MAE, and within-spread proportion on the test samples

Fixing $\lambda_{TS} = 100$, the out-of-sample pricing error metrics confirm that SVCJ improves on both benchmark models in a persistent way. Table 3.3.1 summarizes the average out-of-sample pricing diagnostics separately for BTC, ETH, and the pooled sample. The pattern is consistent across all three metrics: Black performs worst throughout, Heston delivers a large improvement over Black, and SVCJ then improves further on Heston in both assets and in the pooled data.

Table 3.3.1: Average out-of-sample pricing diagnostics at $\lambda_{TS} = 100$, by asset and model.

	Black			Heston			SVCJ		
Asset	RMSE	MAE	Within spread	RMSE	MAE	Within spread	RMSE	MAE	Within spread
BTC	0.005330	0.003565	25.7%	0.002892	0.001482	45.2%	0.002546	0.001004	61.9%
ETH	0.006151	0.003929	32.1%	0.004485	0.002112	49.1%	0.004227	0.001626	63.1%
Pooled	0.005592	0.003772	28.7%	0.003247	0.001771	47.0%	0.002911	0.001304	62.4%

In BTC, the gain from SVCJ is especially clear. Mean test RMSE falls from 0.005330 under Black and 0.002892 under Heston to 0.002546 under SVCJ, while mean test MAE falls from 0.003565 and 0.001482 to 0.001004. ETH shows the same pattern, although the margin over Heston is somewhat smaller: mean test RMSE declines from 0.006151 under Black and 0.004485 under Heston to 0.004227 under SVCJ, and mean test MAE declines from 0.003929 and 0.002112 to 0.001626. The pooled error measures lead to the same conclusion: mean test RMSE falls from 0.005592 under Black to 0.003247 under Heston and then to 0.002911 under SVCJ, while mean test MAE falls from 0.003772 to 0.001771 and then to 0.001304.

The within-spread statistic tells the same story in more economically interpretable terms. In BTC, the mean share of test options priced inside the quoted bid–ask spread rises from only 25.7% under Black and 45.2% under Heston to 61.9% under SVCJ. In ETH, the same proportion rises from 32.1% and 49.1% to 63.1%. Pooled across both assets, SVCJ prices 62.4% of the test options inside the spread, compared with 47.0% for Heston and 28.7% for Black.

Taken together, Table 3.3.1 shows that the improvement from SVCJ is not confined to a single asset or a single error metric. The model dominates Black decisively and remains meaningfully better than Heston.

The snapshot-level paths of the error metrics in the figures below show that the improvement is persistent through time. Figure 3.3.2 shows that in BTC, SVCJ beats Heston on test RMSE in 96.1% of snapshots and on test MAE in 97.9% of snapshots. In ETH, the corresponding shares are 92.5% and 97.4%. The within-spread share in Figure 3.3.3 is also higher for SVCJ in 97.4% of BTC snapshots and 93.8% of ETH snapshots. The Black benchmark is almost never competitive. The more relevant comparison is Heston versus SVCJ, and here the jump extension still wins in a large majority of dates, even though the margin is narrower in ETH than in BTC.

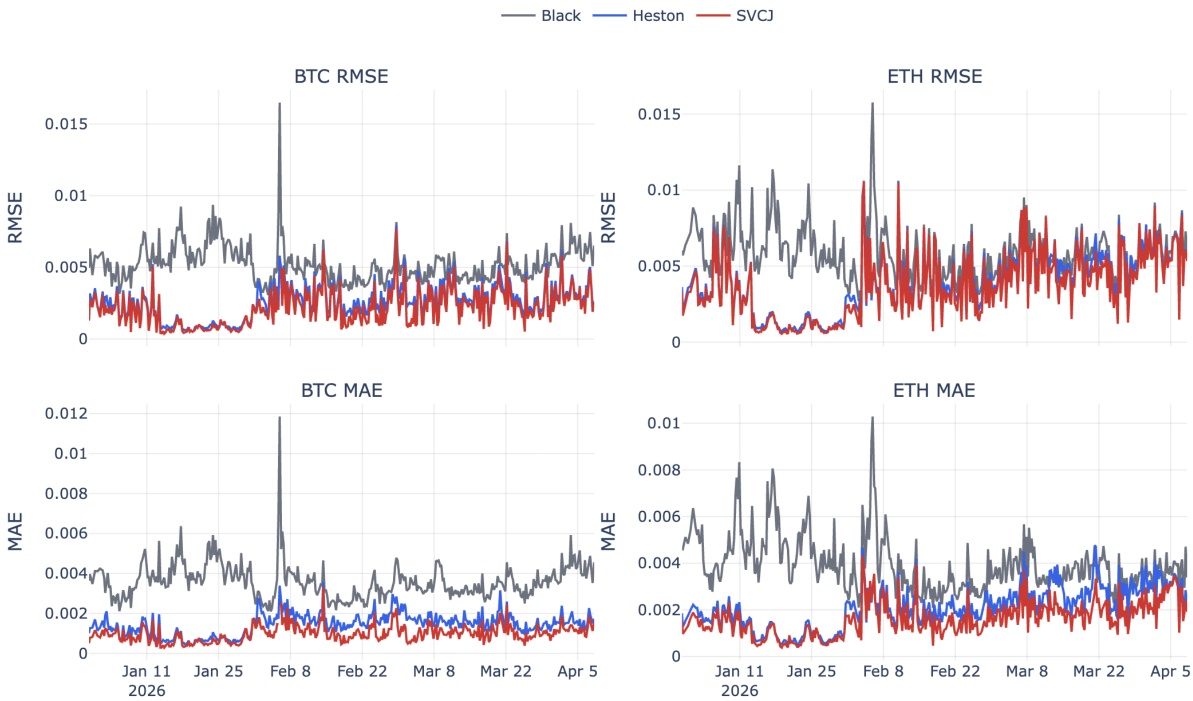

Figure 3.3.2: Snapshot-level out-of-sample RMSE, MAE and within-spread proportion paths at $\lambda_{TS} = 100$.

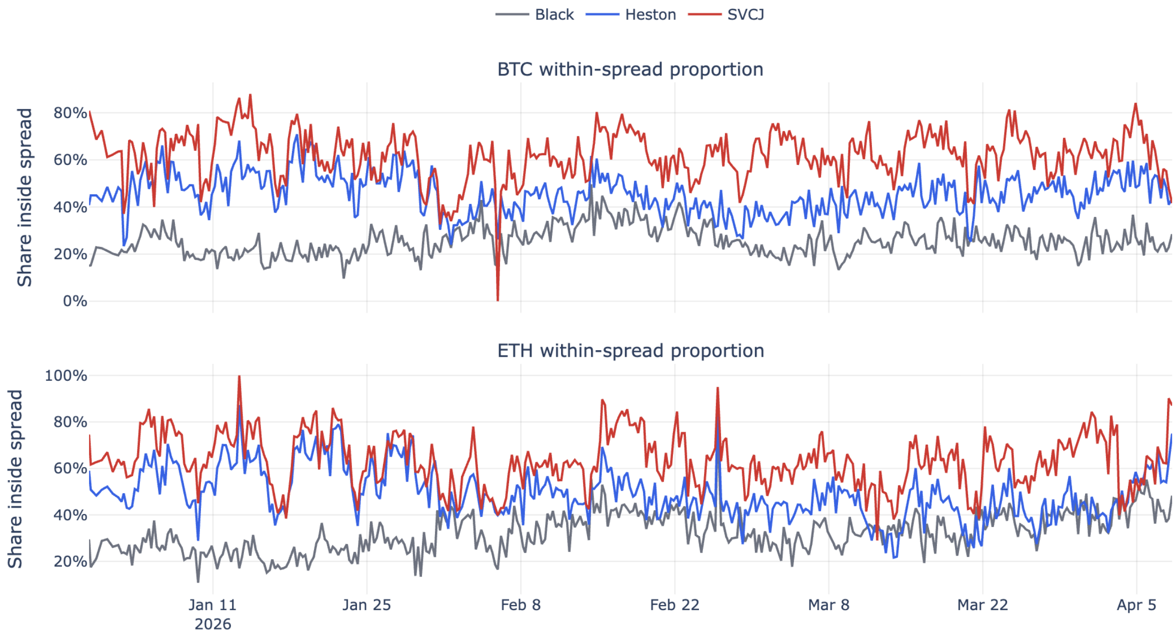

Figure 3.3.3: Snapshot-level within-spread frequencies at $\lambda_{TS} = 100$, shown separately for BTC and ETH.

Two features of the time-series plots are worth emphasizing. First, the SVCJ advantage is not driven by a handful of outliers — the model improves the fit through most of the sample. Second, the gain does not vanish when the surface becomes harder to fit. In the more difficult episodes, where errors are high in all three models, Heston often moves closer to SVCJ, but it does not systematically eliminate the gap. This is exactly the setting in which richer tail and skew dynamics should matter most.

3.3.3 Formal comparison of out-of-sample pricing accuracy

The preceding results compare models using average out-of-sample pricing errors and snapshot-level error paths. To assess whether these differences are statistically significant, we additionally apply Diebold–Mariano tests (Diebold and Mariano, 1995; Harvey et al., 1997) for equal predictive accuracy to the snapshot-level out-of-sample loss series.

For each currency, model m, and snapshot t, define the out-of-sample pricing loss as the mean squared pricing error on the test contracts in that snapshot:

$$L_{m,t} = \frac{1}{N_t} \sum_{i=1}^{N_t} \left(P_{m,t,i}^{model} - P_{t,i}^{mkt} \right)^2 .$$

For a pair of models a and b, define the loss differential

$$d_t^{a,b} = L_{a,t} - L_{b,t}.$$

Thus, $d_t^{a,b} > 0$ means that model b has lower out-of-sample pricing loss than model a. The null hypothesis is

$$H_0 : \mathbb{E}\left[d_t^{a,b}\right] = 0,$$

against the two-sided alternative that the two models have different expected predictive accuracy.

Because option-chain snapshots are collected roughly every six hours, the loss differential series may be serially correlated. The Diebold–Mariano statistic is therefore computed using a Newey–West HAC standard error (Newey and West, 1987) with eight lags, corresponding to two days of six-hour snapshots:

$$DM = \frac{\bar{d}^{a,b}}{\widehat{\text{se}}_{NW}\left(\bar{d}^{a,b}\right)}.$$

Under the null hypothesis of equal predictive accuracy and standard weak-dependence conditions for the loss differential series, the Diebold–Mariano statistic is asymptotically standard normal:

$$DM = \frac{\bar{d}}{\widehat{\text{se}}_{NW}(\bar{d})} \xrightarrow{d} \mathcal{N}(0, 1).$$

Two-sided p-values are therefore computed as

$$p = 2\left[1 - \Phi(|DM|)\right].$$

Table 3.3.2 below reports the sample mean loss differential, the HAC confidence interval for the mean loss differential, the DM statistic, and the two-sided p-value.

Table 3.3.2: Diebold–Mariano tests for equal out-of-sample pricing accuracy at $\lambda_{TS} = 100$.

Currency	Model pair (a, b)	Snapshots	$\bar{d}^{a,b} \times 10^6$	95% CI $\times 10^6$	DM statistic	p-value
BTC	Black, Heston	388	21.287	[17.509, 25.064]	11.05	2.30×10^{-28}
BTC	Heston, SVCJ	387	1.715	[1.391, 2.039]	10.38	3.04×10^{-25}
BTC	Black, SVCJ	387	22.442	[19.095, 25.788]	13.14	1.85×10^{-39}
ETH	Black, Heston	388	18.312	[12.623, 24.000]	6.31	2.80×10^{-10}
ETH	Heston, SVCJ	388	2.189	[1.521, 2.858]	6.42	1.39×10^{-10}
ETH	Black, SVCJ	388	20.501	[15.145, 25.857]	7.50	6.28×10^{-14}

Note: The loss differential is $d_t^{a,b} = L_{a,t} - L_{b,t}$, so positive values imply lower out-of-sample pricing loss for model b.

The formal tests support the ranking obtained from the pricing diagnostics above. For both BTC and ETH, Black is rejected against Heston, Heston is rejected against SVCJ, and Black is rejected against SVCJ. All estimated mean loss differentials are positive, all

HAC confidence intervals exclude zero, and all two-sided p-values are well below conventional significance levels. The evidence therefore indicates that the SVCJ improvement over Heston is not only visible in the average RMSE and MAE statistics, but is also statistically significant at the snapshot level under a HAC-adjusted equal-predictive-accuracy test.

3.3.4 Out-of-sample pricing diagnostics by maturity and moneyness

To check whether the improved out-of-sample pricing performance of the SVCJ model holds up across different maturity and moneyness values, we partition the option-level test sample separately for BTC and ETH into equal-count buckets by maturity and by moneyness.

As shown in Figure 3.3.4, the SVCJ model consistently outperforms the benchmarks across the full surface in both assets. In BTC, SVCJ dominates Heston in every maturity bucket. The gain is already visible in the shortest-expiry bucket, where RMSE falls from 0.001845 under Heston to 0.001702 under SVCJ, but it becomes even clearer further out the term structure. In the longest-maturity bucket, BTC RMSE falls from 0.004210 to 0.003624. The ETH pattern is more modest but still systematic: SVCJ improves on Heston in all five maturity buckets, with RMSE reductions ranging from about 4.7% to 9.2%.

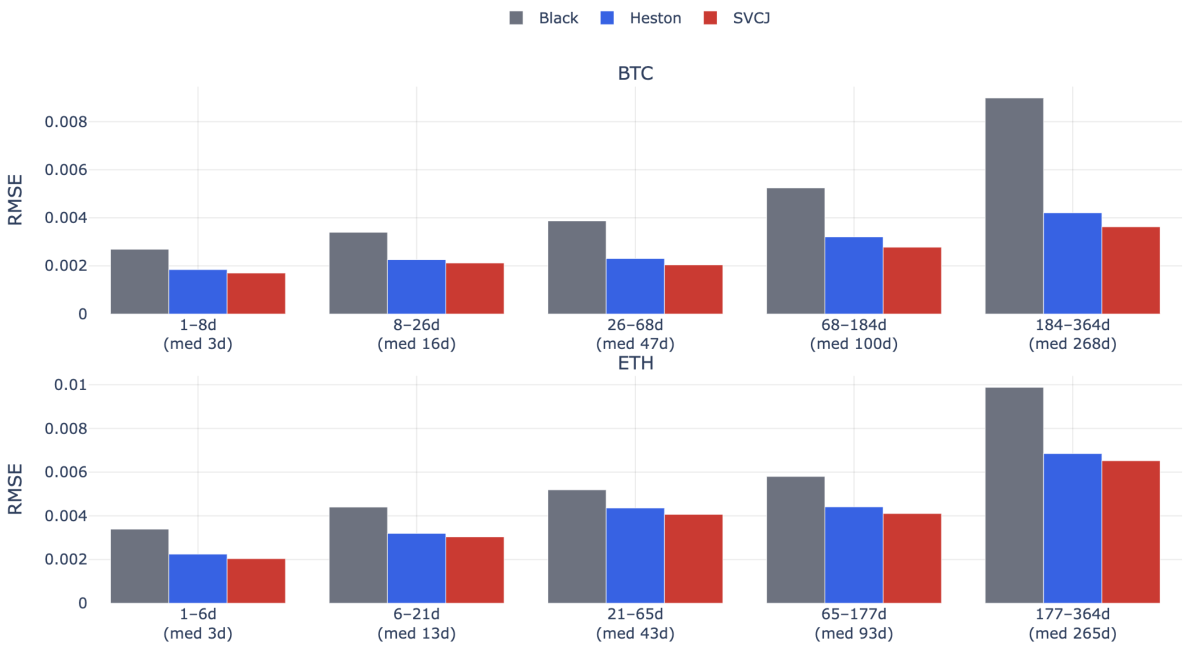

Figure 3.3.4: Out-of-sample mean RMSE by equal-count maturity buckets, separately for BTC and ETH at $\lambda_{TS} = 100$.

The moneyness breakdown in Figure 3.3.5 is even more informative. In BTC, the largest gains occur in the low-moneyness region, where the surface is hardest to match with a pure diffusion model. In the lowest BTC moneyness bucket, test RMSE falls from 0.002797 under Heston to 0.002033 under SVCJ, while the Black benchmark is far worse at 0.008054. The same qualitative result holds in ETH: the lowest moneyness bucket improves from 0.003060 under Heston to 0.002522 under SVCJ. Around the center of the surface, the gains remain positive, but they are less dramatic. At the highest ETH moneyness bucket, for example, Heston and SVCJ are very close in RMSE terms (0.007284 versus 0.007232), which is exactly what one would expect if the incremental value of the jump block is concentrated in the more skewed regions of the volatility surface.

Taken together, these bucket diagnostics support a simple but powerful interpretation: the SVCJ model is not just better on average, it is better precisely in the regions where richer dynamics should matter most.

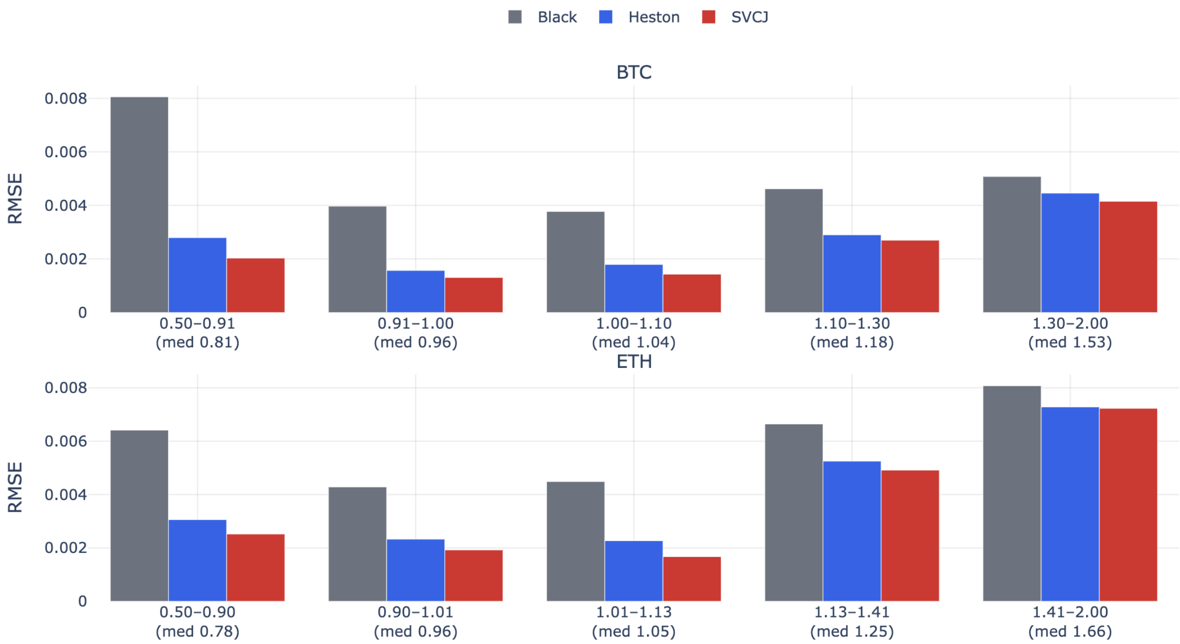

Figure 3.3.5: Out-of-sample mean RMSE by equal-count moneyness buckets, separately for BTC and ETH at $\lambda_{TS} = 100$.

3.3.5 Identification issues and parameter stability

The pricing results are strong, but the parameter paths should not be read as if every component of the SVCJ state vector was strongly identified at every snapshot. Figures 3.3.6 and 3.3.7 at the end of this chapter show selected calibrated parameter paths for the SVCJ model, separately for BTC and ETH. Even after fixing time smoothing at $\lambda_{TS} = 100$, several parameters show unstable paths over time, especially in ETH.

Starting with the more stable parameter paths, BTC loads heavily on a negative diffusion correlation between the returns and volatility processes, with mean $\rho = -0.503$, whereas ETH is close to zero on average, with mean $\rho \approx 0.001$. Economically, this suggests that the BTC surface uses a pronounced leverage-type channel to generate skew, while ETH relies less on diffusion leverage and more on the volatility and jump block. The variance parameters tell a similar story. ETH has a much larger average long-run variance ($\theta = 0.242$ versus 0.105 in BTC), a higher initial variance ($v_0 = 0.386$ versus 0.234), and substantially higher vol-of-vol ($\sigma_v = 2.545$ versus 1.140). In plain terms, ETH requires a steeper and more reactive volatility structure than BTC.

The jump block also differs meaningfully across assets. Both markets favour negative average price jumps: the mean ℓ_y is -0.155 in BTC and -0.102 in ETH. BTC, however, still uses a nontrivial price-jump dispersion, with mean $\sigma_y = 0.167$. ETH is different. There the median σ_y is essentially zero, while the median co-jump dependence parameter is $\rho_j = -0.999$, right at the lower boundary. At the same time, the average jump intensity is higher in ETH ($\lambda = 2.483$) than in BTC ($\lambda = 1.484$). The conclusion is that the ETH surface often prefers to encode jump risk through frequent downside co-jumps in price and variance, rather than through a separate, freely varying jump-size dispersion parameter.

This is also where the identification warning becomes unavoidable. In ETH, $|\rho_j| > 0.95$ in 53.4% of snapshots, and the Feller condition violates in 56.2% of snapshots. That does not invalidate the pricing comparison, but it does mean that some of the ETH jump and variance parameters should be interpreted as weakly identified parameters rather than clean structural estimates. In BTC the same issue is present but milder: $|\rho| > 0.95$ in 11.9% of snapshots, $|\rho_j| > 0.95$ essentially never occurs, and the Feller condition is violated in around 9.0% of snapshots.

The correlation structure of the parameter estimates points to similar conclusions. In BTC, the long-run variance θ and the initial variance v_0 are positively correlated (about 0.49), which is consistent with partial substitution between current and long-run variance levels. In ETH, the jump intensity λ and the jump-size dispersion σ_y are strongly positively correlated (about 0.71), suggesting that the calibration often identifies the overall jump contribution more clearly than its decomposition into jump frequency and jump-size uncertainty.

The economically interpretable part of the parameter dynamics is therefore the broad regime movement, not every single wiggle. The pair (v_0, θ) tracks the volatility regime. The correlation parameter ρ governs how much of the smile asymmetry is generated by diffusion leverage. The parameter σ_v controls how sharply the short end of the surface can bend. The jump-intensity and jump-size parameters, $(\lambda, \ell_y, \sigma_y, \ell_v, \rho_j)$, then pick up the part of the cross section that looks too asymmetric or heavy-tailed to be explained by a pure stochastic-volatility diffusion.

However, it should be noted that the instability of the parameter paths and the identification issues highlighted above warrant a more nuanced approach to parameter estimation in the SVCJ model. While the current cross-sectional calibration methodology may give parameter combinations that perform well in terms of fitting model prices to observed market prices, the large number of parameters of the model leads to many different local minima producing similar pricing results, which is the reason behind the instability and weak identification, making economic interpretation of the parameter paths difficult. A better approach would be to utilise the panel-data structure embedded in the successive option chain snapshots and employ a dynamic calibration method that explicitly accounts for the time-series structure of past calibrated parameter paths. For example, calibration can be formulated as a state-space problem with Kalman/particle filtering methods used to estimate the parameters, which are treated as hidden states. This exercise is left to an extension to the current research.

Figure 3.3.6: Selected SVCJ parameter paths on BTC data

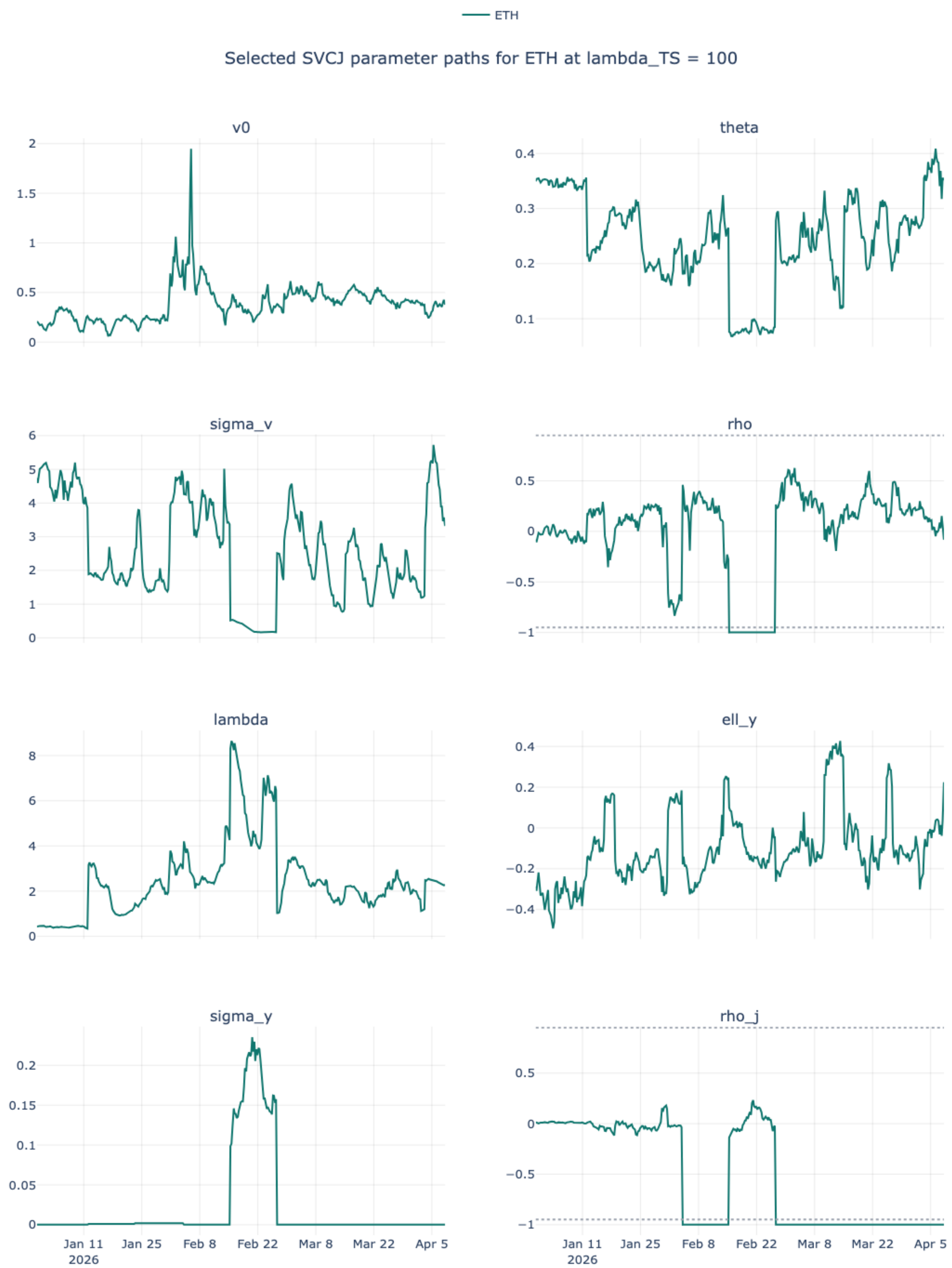

Figure 3.3.7: Selected SVCJ parameter paths on ETH data

4. Delta-Hedging Inverse Options Using the Calibrated Models

This chapter evaluates whether the pricing models studied in Chapters 2 and 3 also deliver economically meaningful improvements in out-of-sample hedging performance. The central questions are (i) whether dynamic delta hedging materially reduces P&L over the unhedged portfolio, (ii) whether the premium-adjusted net delta matters empirically, and (iii) whether the superior pricing fit of the richer models translates into materially better hedge performance.

The main empirical result is that dynamic hedging substantially reduces interval-level P&L relative to leaving the inverse option unhedged; however, once the hedge rule is fixed, the differences between Black, Heston, and SVCJ are economically indistinguishable. The dominant effect is the choice between the plain delta hedge and the premium-adjusted net-delta hedge of Lucic and Sepp (2024). Across both BTC and ETH, the net-delta hedge delivers materially larger reductions in RMSE, MAE, variance, and tail losses than the plain delta hedge.

4.1 Empirical hedging setup

The hedging backtest is constructed as an out-of-sample interval-level hedging P&L analysis on the test option sample. At each observation time t_i, the option is marked using the next available snapshot, and the interim P&L of the corresponding perpetual-futures hedge is computed over the same interval. The goal is to compare the one-step hedging performance of the three pricing models – Black, Heston, and SVCJ – under two hedge rules:

1. the regular-delta hedge based on the conventional futures delta; and
2. the premium-adjusted net-delta hedge appropriate for inverse options.

The backtest therefore addresses three empirical questions:

1. Does dynamic hedging materially reduce the P&L of a short inverse option portfolio over a (roughly) six-hour rebalancing interval?

2. Does the premium adjustment embedded in the net delta empirically matter for hedging performance?

3. Do the pricing gains of the SVCJ model translate into superior hedging performance once the hedge is implemented with perpetual futures rather than the exact dated futures underlying?

4.1.1 Backtest construction

The backtest uses the calibrated parameter estimates from the baseline regularization choice $\lambda_{TS} = 100$ in Chapter 3. For each option in the test sample, the pricing engine produces the model-implied inverse option value $V_{t_i}^{\text{coin}}$ and the corresponding regular delta $\Delta_{t_i}^{\text{reg}}$. Following Sections 2.4 and 2.5, let $F_{t_i}^{(T)}$ denote the dated-futures proxy associated with the option chain at time t_i, and let $F_{t_i}^{\perp}$ denote the perpetual futures mark price used for hedging. Because the hedge is executed in inverse perpetuals, the implementation expresses the hedge size as a USD notional rather than as a contract count. The hedge ratios are constructed as follows:

$$\Delta_{t_i}^{\text{reg}} = \frac{\partial V_{t_i}^{\text{USD}}}{\partial F_{t_i}^{(T)}}, \qquad \Delta_{t_i}^{\text{net}} = \Delta_{t_i}^{\text{reg}} - V_{t_i}^{\text{coin}}.$$

For a detailed discussion of how $\Delta_{t_i}^{\text{reg}}$ is calculated under the Black, Heston and SVCJ models, see Appendix D.

For the empirically relevant case of a short option position hedged using perps, the corresponding hedge notionals are therefore

$$N_{\text{USD},t_i}^{\text{reg}} = F_{t_i}^{\perp}\, \Delta_{t_i}^{\text{reg}}, \qquad N_{\text{USD},t_i}^{\text{net}} = F_{t_i}^{\perp}\, \Delta_{t_i}^{\text{net}}.$$

The realized P&L of the position in the perp is then

$$\Pi^{\perp}(t_i, t_{i+1}; N_{\text{USD},t_i}^{(m)}) = N_{\text{USD},t_i}^{(m)} \left(\frac{1}{F_{t_i}^{\perp}} - \frac{1}{F_{t_{i+1}}^{\perp}} \right), \qquad m \in \{\text{reg}, \text{net}\}.$$

Several implementation choices matter for interpretation. First, the hedge is rebalanced on the snapshot grid, and in the final sample the median interval length is approximately 6.01 hours for both BTC and ETH. Second, only intervals with a valid next option quote and valid perpetual prices at both ends of the interval are retained. Third, intervals that cross option expiry are excluded. Finally, funding effects and transaction costs are excluded from hedged P&L metrics reported in this chapter. The reported results should therefore be read as clean delta-hedging diagnostics rather than fully frictional trading P&L.

4.1.2 Evaluation metrics

Hedging performance is evaluated using several complementary diagnostics. Let $\varepsilon^{unh}_{t_i,t_{i+1}}$ denote the one-interval P&L of the unhedged short inverse option position, and let $\varepsilon^{hed}_{t_i,t_{i+1}}$ denote the corresponding hedged P&L. The main metrics are:

1. the reduction in RMSE and MAE,
$$\text{RMSE reduction} = 1 - \frac{\text{RMSE}(\varepsilon^{hed})}{\text{RMSE}(\varepsilon^{unh})}, \qquad \text{MAE reduction} = 1 - \frac{\text{MAE}(\varepsilon^{hed})}{\text{MAE}(\varepsilon^{unh})};$$
2. the reduction in the variance of interval P&L,
$$\text{Variance reduction} = 1 - \frac{\widehat{\text{Var}}(\varepsilon^{hed})}{\widehat{\text{Var}}(\varepsilon^{unh})};$$
3. left-tail diagnostics, namely the empirical 5% quantile and 5% expected shortfall of hedged interval P&L;
4. the hit rate of absolute improvement, i.e. the share of intervals for which the hedged absolute P&L is strictly smaller than the unhedged absolute P&L,
$$\text{Hit rate} = \frac{1}{n}\sum_i \mathbf{1}\left\{|\varepsilon_i^{hed}| < |\varepsilon_i^{unh}|\right\}.$$

Taken together, these measures distinguish between average hedging quality, dispersion reduction, and downside-risk control. This matters because a hedge can improve RMSE while still leaving the left tail relatively poor, or vice versa.

4.2 Hedging sample and descriptive statistics

Table 4.2.1 summarizes the final out-of-sample hedging sample. The starting point is the test option panel produced by the calibration workflow. An interval is retained if the next option quote is available, the perpetual futures quote is available at both ends of the interval, and the interval does not cross expiry. The resulting eligible shares are high in both markets: 99.17% for BTC and 99.11% for ETH.

Table 4.2.1: Final out-of-sample hedging sample construction.

Currency	Raw intervals	Eligible intervals	Eligible share	Unique instruments	Unique snapshots	Median intervals per snapshot	Median Δt (h)	Cheap-option share
BTC	57,494	57,014	99.17%	3,539	388	144.5	6.01	15.63%
ETH	49,532	49,091	99.11%	3,666	388	129.0	6.01	11.52%

Two features of Table 4.2.1 are worth emphasizing. First, the sample remains dense at the snapshot level: the median test snapshot contributes 144.5 BTC intervals and 129.0 ETH intervals. Second, the cheap-option share is non-trivial, especially in BTC. Cheap options are defined (somewhat arbitrarily) as options with prices less than 0.005 coin units, for both BTC and ETH. This threshold is about 3-4% of the mean BTC and ETH option premia in the test sample. This matters because very low-premium options can mechanically inflate percentage improvements in some settings. Percentage reduction metrics are normalized by the unhedged P&L scale, so small absolute changes in very low-premium options can translate into disproportionately large relative improvements. This issue is examined in Section 4.5.

4.3 Headline out-of-sample hedging results

4.3.1 Pooled comparison across models and hedge rules

Table 4.3.1 reports the mean interval hedging error metrics for the full test sample. The table already reveals the main result of the chapter: hedging helps a great deal, and the net-delta adjustment matters far more than the model choice between Black, Heston, and SVCJ.

Table 4.3.1: Headline out-of-sample hedging results: error and variance reduction on the full test sample.

Currency	Model	Hedge	RMSE reduction	MAE reduction	Variance reduction	Hit rate of abs. improvement
BTC	Black	Delta	52.8%	60.7%	77.7%	77.7%
BTC	Black	Net delta	65.2%	77.2%	87.9%	79.4%
BTC	Heston	Delta	52.3%	59.5%	77.2%	78.2%
BTC	Heston	Net delta	64.9%	76.8%	87.7%	79.7%
BTC	SVCJ	Delta	52.2%	59.4%	77.2%	78.5%
BTC	SVCJ	Net delta	64.8%	76.8%	87.6%	80.1%
ETH	Black	Delta	48.1%	54.9%	73.0%	77.2%
ETH	Black	Net delta	62.6%	73.5%	86.0%	79.4%
ETH	Heston	Delta	47.9%	54.3%	72.8%	77.4%
ETH	Heston	Net delta	62.5%	73.5%	86.0%	79.8%
ETH	SVCJ	Delta	47.9%	54.3%	72.8%	77.6%
ETH	SVCJ	Net delta	62.5%	73.5%	85.9%	80.0%

In BTC, the plain delta hedge reduces RMSE by roughly 52.2–52.8% across the three models, whereas the net-delta hedge increases the reduction to roughly 64.8–65.2%. The same pattern appears in ETH: the plain delta hedge reduces RMSE by roughly 47.9–48.1%, while the net-delta hedge raises this to 62.5–62.6%. The ranges across models are tiny. Measured directly, the within-rule cross-model spread in RMSE reduction is only 0.56 percentage points for BTC delta, 0.31 percentage points for BTC net delta, 0.20 percentage points for ETH delta, and 0.07 percentage points for ETH net delta. The gain from switching from Black to SVCJ under a fixed hedge rule is therefore economically negligible compared with the gain from switching from delta to net delta as the hedge ratio.

The MAE and variance results point in exactly the same direction. For BTC, net delta raises the MAE reduction from about 59–61% to about 77% and the variance reduction from about 77% to about 88%. For ETH, net delta raises the MAE reduction from about 54–55% to about 73–74%, and the variance reduction from about 73% to about 86%. The conclusion is that over these six-hour intervals, the empirical value of the hedging framework comes primarily from using the inverse-option-consistent hedge ratio, not from using the most flexible pricing model.

Figures 4.3.1 and 4.3.2 below summarise the results presented above graphically by showing the test sample mean interval-level PnL RMSE/MAE reduction vs the unhedged position, by model and hedge ratio.

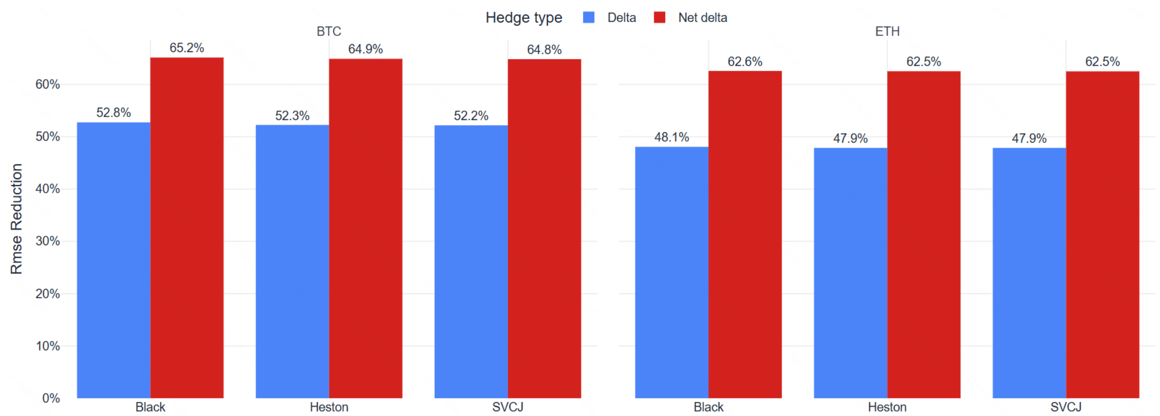

Figure 4.3.1: Test sample mean interval-level P&L RMSE reduction vs the unhedged position, by model and hedge ratio.

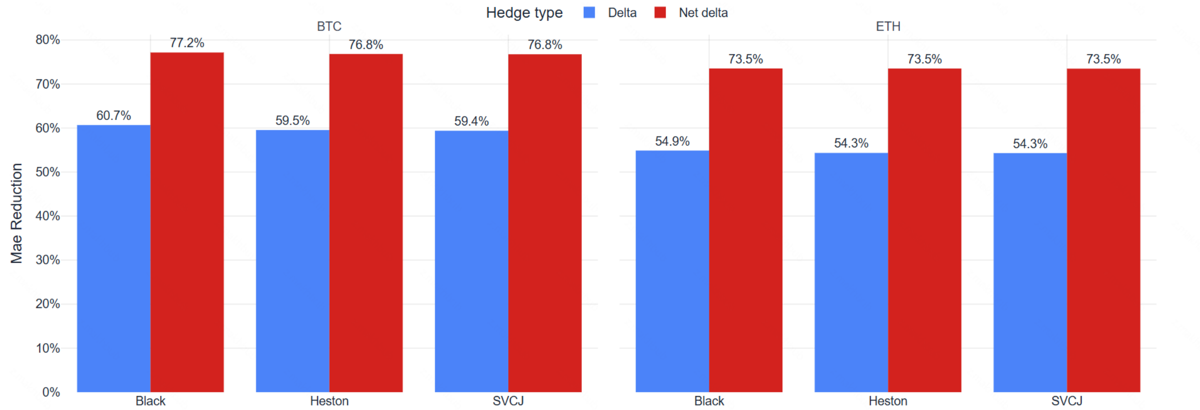

Figure 4.3.2: Test sample mean interval-level P&L MAE reduction vs the unhedged position, by model and hedge ratio.

4.3.2 Tail-risk evidence

The same conclusion emerges from the left-tail diagnostics in Table 4.3.2. For BTC, the 5% expected shortfall of hedged interval P&L improves from about -0.0113 coin under the plain Black delta hedge to -0.0068 coin under the Black net-delta hedge. ETH shows a similarly large improvement, from about -0.0186 coin to about -0.0119 coin. Again, the cross-model differences within a fixed hedge rule are minute. What moves the tail risk is the premium adjustment embedded in the net delta.

Table 4.3.2: Mean out-of-sample tail-risk diagnostics on the full test sample.

Currency	Model	Hedge	5% quantile of hedged P&L	5% expected shortfall
BTC	Black	Delta	-0.0051	-0.0113
BTC	Black	Net delta	-0.0023	-0.0068
BTC	Heston	Delta	-0.0053	-0.0115
BTC	Heston	Net delta	-0.0024	-0.0070
BTC	SVCJ	Delta	-0.0054	-0.0115
BTC	SVCJ	Net delta	-0.0024	-0.0070
ETH	Black	Delta	-0.0090	-0.0186
ETH	Black	Net delta	-0.0042	-0.0119
ETH	Heston	Delta	-0.0091	-0.0188
ETH	Heston	Net delta	-0.0042	-0.0120
ETH	SVCJ	Delta	-0.0092	-0.0188
ETH	SVCJ	Net delta	-0.0042	-0.0120

4.3.3 Snapshot-level stability

Because the pooled results compress the entire sample into a few averages, it is useful to inspect the time-series stability of the hedging gains. Since the cross-model hedge differences are negligible in the pooled results, the stability diagnostic is shown for Black as a representative benchmark. This keeps the exposition focused on the hedge-rule comparison rather than repeating near-identical plots for all three models.

Figures 4.3.3 and 4.3.4 show the snapshot level RMSE reductions of the hedged porfolios. The key result from the previous section remains unchanged: the net-delta hedge dominates the plain delta hedge through most of the sample. For Black, net delta delivers a higher snapshot-level RMSE reduction than plain delta in 84.2% of BTC snapshots and 80.6% of ETH snapshots. An important result is that using the plain delta as the hedge ratio leads to negative RMSE reductions on some snapshots, that is, the plain delta-hedged portfolio leads to larger P&L than the unhedged portfolio. This is especially visible in Figure 4.3.4, and reinforces the conclusion that the premium net-delta is the proper hedge ratio when hedging inverse options.

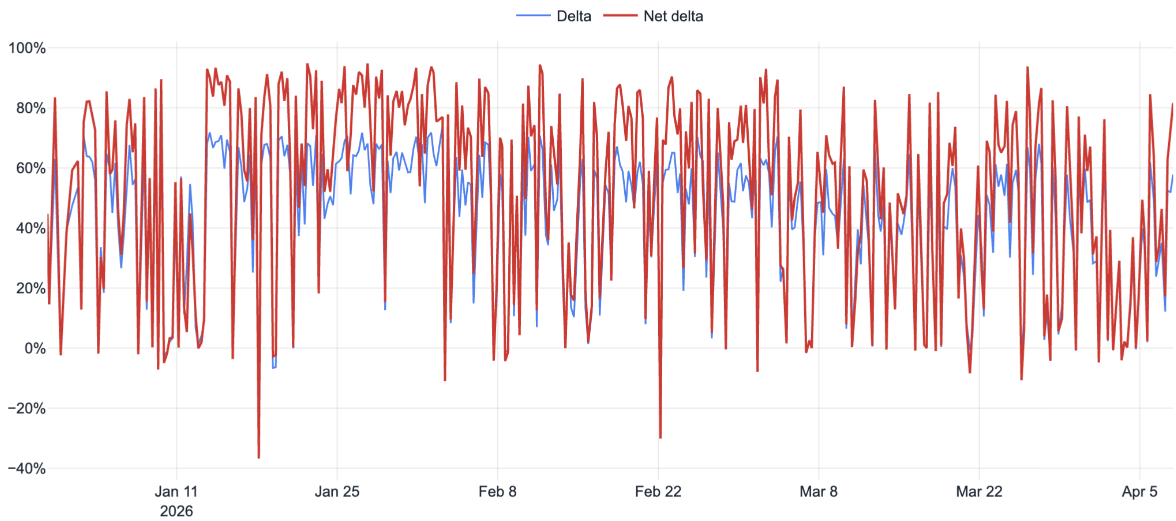

Figure 4.3.3: Snapshot-level BTC portfolio RMSE reduction for the representative Black model.

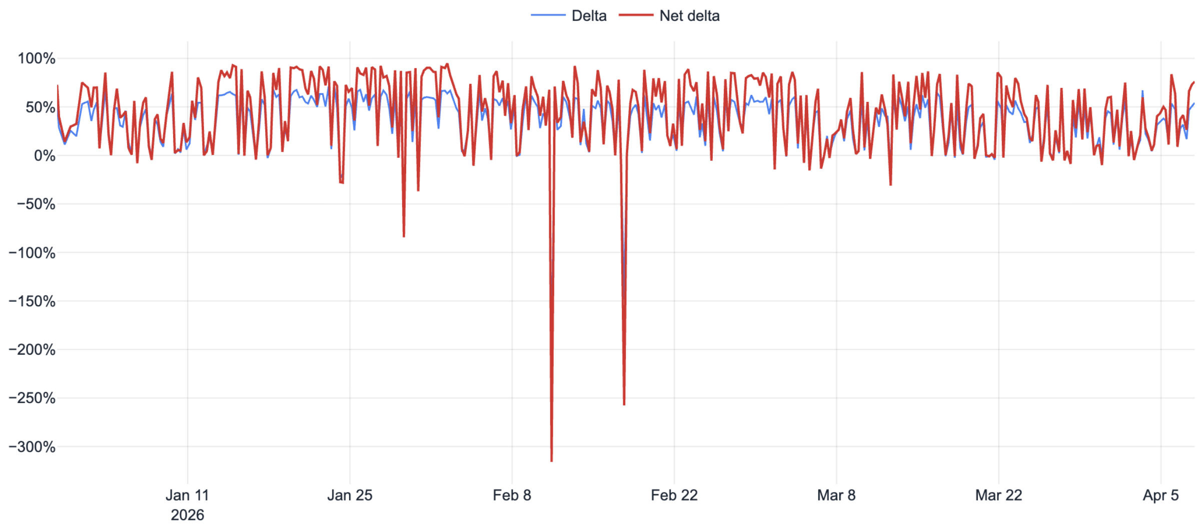

Figure 4.3.4: Snapshot-level ETH portfolio RMSE reduction for the representative Black model.

4.3.4 Diebold-Mariano test for regular-delta vs net-delta hedges

The preceding results show that the premium-adjusted net-delta hedge dominates the regular-delta hedge in the pooled error metrics and in most snapshot-level RMSE comparisons. To test this difference formally, we apply the same Diebold-Mariano test logic as in Section 3.3.3 to the snapshot-level hedging loss series under the representative Black model.

For each currency, snapshot t, and hedge rule $h \in \{\Delta, \Delta^{net}\}$, define the snapshot-level hedging loss as the mean squared hedged P&L on the eligible test intervals in that snapshot:

$$L_{h,t}^{Black} = \frac{1}{N_t} \sum_{i=1}^{N_t} \left(\varepsilon_{t,i}^{Black,h} \right)^2,$$

where $\varepsilon_{t,i}^{Black,h}$ is the interval-level hedged P&L from using hedge rule h. The loss differential is

$$d_t^{\Delta,net} = L_{\Delta,t}^{Black} - L_{\Delta^{net},t}^{Black}.$$

Thus, $d_t^{\Delta,net} > 0$ means that the net-delta hedge has lower mean squared hedging error than the regular-delta hedge. The null hypothesis is

$$H_0 : \mathbb{E}\left[d_t^{\Delta,net} \right] = 0.$$

As in the pricing comparison, the standard error of the mean loss differential is computed using a Newey–West HAC estimator with eight lags, corresponding to two days of six-hour snapshot intervals.

Table 4.3.3: Diebold–Mariano tests comparing regular-delta and net-delta hedges under the representative Black model.

Currency	Snapshots	$\bar{d}^{\Delta,net} \times 10^6$	95% CI $\times 10^6$	DM statistic	p-value
BTC	387	8.858	[5.503, 12.214]	5.17	2.29×10^{-7}
ETH	387	24.117	[16.920, 31.314]	6.57	5.10×10^{-11}

Note: The loss differential is $d_t^{\Delta,net} = L_{\Delta,t}^{Black} - L_{\Delta^{net},t}^{Black}$, so positive values imply lower snapshot-level mean squared hedging error for the premium-adjusted net-delta hedge.

The results confirm the conclusions from the preceding sections. For both BTC and ETH, the estimated mean loss differential is positive and statistically significant, and the HAC confidence intervals exclude zero. The null of equal hedging accuracy between the regular-delta and net-delta hedge is therefore rejected for both currencies. This supports the interpretation that the improvement from the net-delta hedge is not only a point-estimate feature of the pooled sample, but a statistically persistent reduction in snapshot-level mean squared hedging error.

4.4 Cross-sectional diagnostics

The pooled results show that hedging works and that net delta matters. The next question is where the hedge works best and where it struggles. As before, the cross-sectional bucket figures are shown for the representative Black specification.

4.4.1 By maturity bucket

Figure 4.4.1 groups the representative-model test sample into within-currency equal-count maturity buckets. The maturity pattern is informative but not perfectly monotone. In BTC, the Black net-delta hedge performs best in the short-to-intermediate region, with RMSE reductions of 69.3%, 70.4%, and 73.3% in the first three buckets, then drops sharply to 52.0% in the 68–184 day bucket, and partially recovers to 66.5% in the longest bucket. ETH shows a cleaner deterioration with maturity: RMSE reduction declines from 69.8% in the 1–6 day bucket to 52.4% in the 177–365 day bucket, with the main break occurring beyond roughly two to three months. The common message is still clear: longer-dated options are harder to hedge over short intervals because more of their risk comes from changes in the shape of the surface, residual basis variation, and model misspecification.

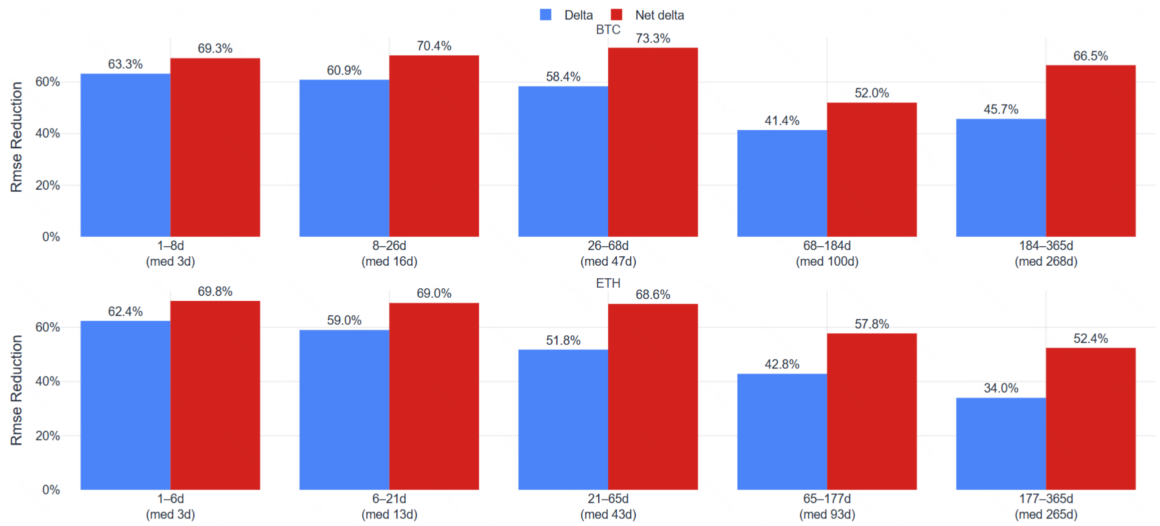

Figure 4.4.1: RMSE reduction by equal-count maturity bucket for the representative Black model.

4.4.2 By moneyness bucket

Figure 4.4.2 reports the same exercise by within-currency equal-count moneyness buckets. The moneyness pattern is clearer than the maturity pattern. Hedging is strongest around the centre (close to ATM) of the surface and weaker in the wings. In BTC, the Black net-delta hedge reduces RMSE by 69.7% in the near-ATM bucket 0.91–1.00 and 68.9% in the 1.00–1.10 bucket, but only 59.9% in the low-moneyness bucket 0.50–0.91 and 62.8% in the far right tail 1.30–2.00. ETH shows the same broad pattern, although asymmetrically. The strongest bucket is the central 1.01–1.13 range, where RMSE reduction reaches 71.0%, while the left wing is clearly hardest at 50.5%.

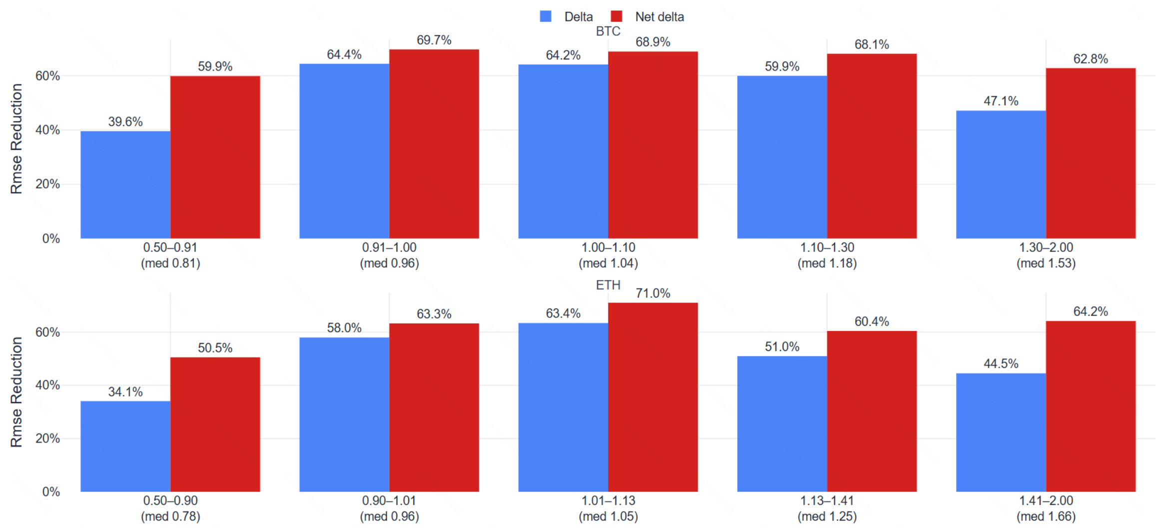

Figure 4.4.2: RMSE reduction by equal-count moneyness bucket for the representative Black model.

This result lines up with the pricing evidence from Chapter 3. The wings are precisely where the option surface is harder to fit and where inverse-option payoffs are more non-linear in practice. It is therefore unsurprising that the residual hedging error is larger there.

4.4.3 By basis bucket

Figure 4.4.3 repeats the same analysis by basis buckets, where the basis is defined as the absolute gap between the price of the perpetual used for hedging and the underlying dated futures price associated with the option.

The economic interpretation of the results is that as the basis widens, hedging becomes materially harder. In BTC, the Black net-delta hedge reduces RMSE by 71.9% in the second-lowest basis bucket (0.04–0.18%, median 0.08%), but only 51.2% in the highest basis bucket (1.72–4.72%, median 2.70%). ETH shows an even steeper deterioration: net-delta RMSE reduction is close to 70% in the two lowest basis buckets, but falls to only 46.9% in the highest basis bucket (1.61–4.38%, median 2.51%). The relationship is not perfectly monotone in the middle buckets, but the high-basis regime is unequivocally worse for both assets.

This explains why the richer pricing model does not automatically produce much better hedging outcomes in this setup. Once the hedge is implemented with perpetual futures rather than with the exact dated futures underlying, basis risk becomes a major determinant of residual P&L. Fine differences in model-implied delta are then largely insignificant in comparison in a first-order delta hedge.

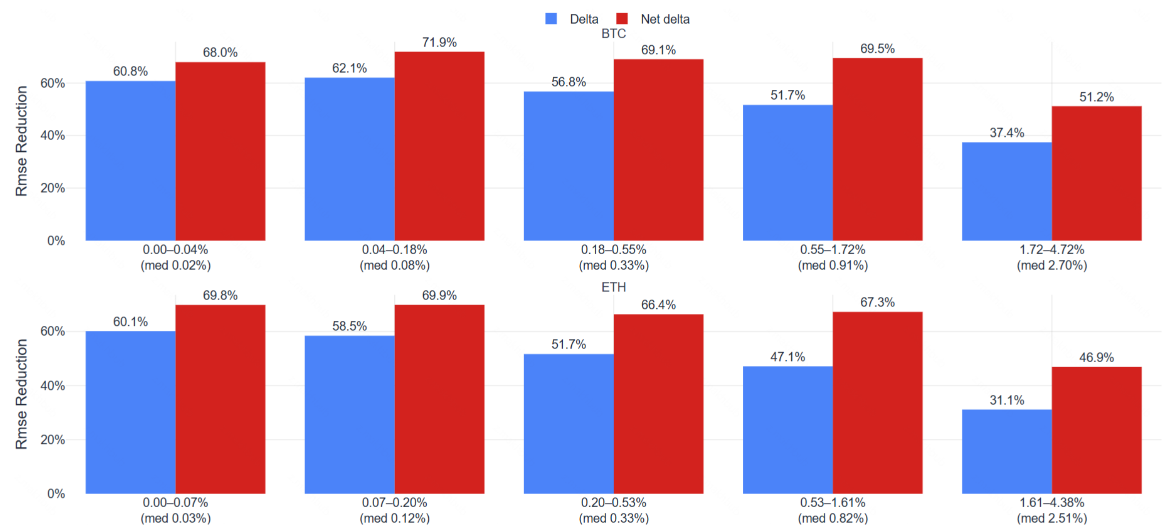

Figure 4.4.3: RMSE reduction by equal-count absolute basis bucket for the representative Black model.

4.5 Robustness checks

As explained in Section 4.2, very cheap options can artificially inflate our hedging error metrics, and Section 4.4 showed that hedging results are strongest for shorter maturity options. For this reason, table 4.5.1 reports two simple robustness checks for the representative Black model: excluding options with maturity up to seven days, and excluding both those very short-dated options and cheap options. The qualitative result is unchanged. In every case, net delta continues to dominate plain delta in both BTC and ETH.

The absolute values shift slightly, which is expected. Removing very short-dated options makes the exercise somewhat harder because the easiest near-expiry contracts disappear. Removing cheap options also changes the composition of the sample. But none of this alters the ranking. The premium-adjusted hedge remains clearly superior.

Table 4.5.1: Sensitivity checks for the representative Black specification.

Sample	Currency	Hedge	Intervals	RMSE reduction	MAE reduction	Variance reduction	Hit rate of abs. improvement
Full	BTC	Delta	57,014	52.8%	60.7%	77.7%	77.7%
Full	BTC	Net delta	57,014	65.2%	77.2%	87.9%	79.4%
Full	ETH	Delta	49,091	48.1%	54.9%	73.0%	77.2%
Full	ETH	Net delta	49,091	62.6%	73.5%	86.0%	79.4%
Exclude $\leq$ 7d	BTC	Delta	46,578	51.0%	59.1%	76.0%	77.8%
Exclude $\leq$ 7d	BTC	Net delta	46,578	64.4%	77.7%	87.3%	79.8%
Exclude $\leq$ 7d	ETH	Delta	38,716	45.7%	52.5%	70.6%	77.7%
Exclude $\leq$ 7d	ETH	Net delta	38,716	61.3%	73.9%	85.0%	80.4%
Exclude $\leq$ 7d + cheap	BTC	Delta	41,291	51.0%	59.4%	76.0%	81.3%
Exclude $\leq$ 7d + cheap	BTC	Net delta	41,291	64.4%	78.1%	87.3%	83.5%
Exclude $\leq$ 7d + cheap	ETH	Delta	35,824	45.7%	52.6%	70.6%	79.1%
Exclude $\leq$ 7d + cheap	ETH	Net delta	35,824	61.3%	74.1%	85.0%	82.0%

4.6 Interpretation and link back to the pricing results

The hedging results should be interpreted together with the pricing evidence in Chapter 3. In the calibration exercise, the SVCJ model delivered a clear and persistent improvement in cross-sectional pricing accuracy relative to both Black and Heston. In the hedging exercise, however, this pricing advantage does not translate into a material improvement in short-horizon delta-hedging performance. This is not a contradiction. Pricing and hedging are related but distinct empirical tasks: the former asks which model best fits the option surface at a given snapshot, while the latter asks which hedge rule best reduces one-interval P&L risk when the hedge is implemented with an imperfect delta-one instrument. Two reasons explain why the pricing advantages of the SVCJ model do not translate into hedging advantages.

4.6.1 Basis risk as a dominant source of residual hedging error

A first explanation is that the hedging problem studied in this chapter contains an important instrument mismatch. The option is written on a dated futures contract, whereas the hedge is implemented with the more liquid inverse perpetual futures contract. As a result, even a correctly specified model delta with respect to the dated futures price cannot eliminate the residual P&L generated by movements in the perpetual–term basis. The basis-bucket diagnostics reinforce this interpretation: hedging performance deteriorates materially when the absolute basis between the perpetual and the dated futures proxy becomes large. In such states, the main obstacle is not that Black, Heston, and SVCJ produce radically different deltas, but that all three deltas are applied through a hedge instrument whose price does not move one-for-one with the option underlying. Once this basis risk becomes first-order, fine differences in model-implied delta are naturally pushed into the background.

4.6.2 Higher-order hedges

A second explanation is that the hedging exercise in this chapter is strictly first-order. The comparison is driven only by differences in the model-implied delta, rather than by differences in higher-order sensitivities to the underlying price, variance state, jump risk, or changes in the local shape of the volatility surface. This matters because the main pricing advantage of Heston and especially SVCJ comes from their ability to fit features such as smile curvature, tail asymmetry, and short-end distortions. These features are economically important for valuation, but they need not imply materially different *first-order* delta hedges over short rebalancing intervals.

The incremental value of the richer model may therefore become more visible in hedging problems that go beyond simple delta exposure control. If the hedger also manages gamma, vega, or more general surface risk, then a better fit to the wings and to the curvature of the option surface should matter more directly. In that setting, SVCJ's superior pricing performance could translate into better hedge construction, because the model would be used not only to compute a local first derivative with respect to the underlying, but also to measure and control higher-order exposures. Such an exercise lies beyond the scope of the present chapter, but it is a natural extension of the empirical findings.

Overall, the results imply a sharper interpretation of the thesis evidence. SVCJ is the strongest model for cross-sectional pricing, especially in the regions of the option surface where stochastic volatility and jumps should matter most. For the short-horizon hedging problem considered here, however, the dominant margin of improvement is not model complexity but the use of the inverse-option-consistent premium-adjusted net delta. Better pricing and better first-order hedging are therefore not equivalent criteria in this setting, especially when the hedge is implemented with perpetual futures and is exposed to residual basis risk.

Conclusion

This thesis examined the pricing and hedging of Deribit inverse options under affine stochastic volatility models with jumps. Inverse cryptocurrency options differ in economically meaningful ways from standard cash-settled contracts: the premium and payoff are denominated in coin, the option is written on a dated futures contract, and the practically relevant hedge is implemented with inverse perpetual futures. These market conventions make numeraire choice, contract denomination, and hedge construction central to the analysis. The literature already made clear that two ingredients are indispensable. First, the inverse-option literature shows that inverse contracts should be priced and hedged through a futures-numeraire framework and that the relevant hedge ratio is a premium-adjusted net delta rather than the conventional delta. Second, the crypto option-pricing literature shows that stochastic volatility and jumps are core features of the return dynamics that shape observed smiles, skews, and tail behaviour. The contribution of this thesis was to bring these two strands together in a single empirical framework tailored to Deribit inverse options and to evaluate that framework using both pricing and hedging criteria. The research question was whether a richer crypto-specific affine model could improve the valuation and risk management of inverse options relative to benchmark models, and the working hypotheses were that SVCJ should improve cross-sectional pricing and hedging accuracy and that inverse-option-consistent hedging should outperform plain delta hedging.

The methodology of the thesis was deliberately constructed to connect theory, implementation, and empirical testing. On the theoretical side, the analysis began by adapting the SVCJ model of Hou et al. (2020) to the case in which the underlying is a dated futures contract and the natural pricing measure is the futures-price numeraire. Under that choice of numeraire, the regular USD-like option value can be computed using characteristic-function methods and then converted into the coin-denominated inverse option value by scaling with the current futures price. The affine structure of the model makes this approach tractable: once the characteristic function is available, European option prices can be obtained efficiently through Fourier inversion, and in the empirical implementation this is done with the Carr–Madan FFT procedure. The same framework also yields the distinction between the regular delta and the premium-adjusted net delta, the latter being the hedge ratio that is consistent with the inverse denomination of the option.

On the empirical side, the thesis used a large panel of BTC and ETH Deribit option-chain snapshots collected over time and supplemented these data with perpetual-futures quotes for the hedging exercise. The raw data were filtered using maturity, quote-quality, spread, open-interest, vega, and moneyness criteria in order to retain reasonably liquid quotes and remove obviously problematic observations. Calibration was then performed snapshot by snapshot for the Black, Heston, and SVCJ models using a weighted nonlinear least-squares objective in coin-denominated prices. The weighting scheme downweighted wide markets, while the optimization also incorporated a time-smoothing penalty so that the estimated parameter paths would not fluctuate implausibly from one snapshot to the next in a high-dimensional and partially ill-posed problem. This feature is important, because repeated cross-sectional calibration of flexible models with large numbers of parameters can otherwise generate noisy parameter series that are difficult to interpret. The thesis therefore treated temporal smoothing as a stabilization device, not as a structural claim about latent parameter dynamics, and compared several levels of regularisation before selecting the baseline value.

The hedging methodology added a second layer of realism. Rather than asking whether inverse options can be replicated perfectly in the idealized case of continuous trading in the exact underlying futures contract, the thesis asked whether model-based delta hedging reduces interval risk when the hedge is actually executed with the more liquid inverse perpetual futures contract, and whether the pricing gains of the SVCJ model transfer to hedging gains in practice. The backtest was performed out-of-sample on interval-level P&L, using the next available snapshot to mark the option and the contemporaneous perpetual-futures price change to compute hedge P&L. Two hedge rules were compared for each pricing model: the regular-delta hedge and the premium-adjusted net-delta hedge. Performance was then evaluated by a family of diagnostic metrics, including RMSE and MAE reduction relative to the unhedged position, variance reduction, hit rates of absolute improvement, and left-tail measures such as the 5% quantile and 5% expected shortfall of hedged interval P&L. This structure made it possible to separate model fit from hedge construction and to determine whether the pricing gains of richer models survive the constraints of actual hedge implementation.

The empirical results of the thesis are clear. On the pricing side, the SVCJ model improved on both benchmark models in a persistent and economically meaningful way. Black performed worst throughout, which is unsurprising given the shape of crypto option surfaces, while Heston produced a large improvement over Black by allowing stochastic volatility. SVCJ then improved further by introducing a richer co-jump structure in the returns and volatility processes. This ranking held in both BTC and ETH and was visible across multiple diagnostics, including RMSE, MAE, and the proportion of options priced inside the bid–ask spread. The gains were especially pronounced in the parts of the surface where one would most expect jumps and asymmetric tail behaviour to matter, such as low-moneyness regions and more difficult maturity buckets. In other words, SVCJ was not merely marginally better on average; it was better precisely where a crypto-specific model with richer dynamics should have an advantage.

At the same time, the thesis also showed that stronger pricing performance should not automatically be equated with stronger hedging performance. Dynamic delta-hedging itself produced large gains relative to leaving the short inverse option unhedged, however, once the hedge rule was fixed, the differences between Black, Heston, and SVCJ were economically very small over the six-hour rebalancing intervals considered in the backtest. The dominant empirical effect was not the choice between benchmark and richer pricing models, but the choice between the regular delta and the premium-adjusted net delta. Across both BTC and ETH, the net-delta hedge produced materially larger reductions in RMSE, MAE, and variance, and materially better tail-risk outcomes, than the plain delta hedge. The conclusion is straightforward: for inverse options, getting the contract-specific hedge ratio right matters far more for short-horizon risk reduction than moving from Black or Heston to a richer jump-diffusion specification, at least when hedging is implemented with perpetual futures and therefore exposed to basis risk.

This result does not weaken the contribution of the richer model. A model can produce a markedly better representation of the cross-sectional surface without delivering a correspondingly large improvement in first-order delta hedging over moderate rebalancing intervals. In the present setting, this is plausible for several reasons. First, the hedge is implemented in perpetual futures rather than in the exact dated-futures underlying, so basis risk limits the extent to which pricing gains can be translated into hedge gains. Second, the hedge implemented is only a first-order delta hedge. The pricing gains of SVCJ mainly reflect its ability to fit smile curvature, wings, and maturity effects; these dimensions may matter more directly in delta–gamma hedging, vega hedging, or quadratic hedging than in a one-period delta hedge. In that sense, the results of the thesis are internally coherent: SVCJ is the strongest model for representing the option surface, while the net delta is the relevant hedging ratio for economically meaningful short-horizon first-order delta-hedging of inverse options.

The thesis also highlights several limitations that help define the next research steps. One limitation is that some parameters of the SVCJ model, especially in the ETH sample, are only weakly identified in repeated cross-sectional calibration. This does not invalidate the pricing comparison, but it does mean that individual parameter paths are difficult to interpret economically and that some of the jump and variance parameters are better viewed as effective fit devices than as clean structural estimates at every snapshot. Another limitation is that the hedging exercise focused on first-order exposure only. That is appropriate for establishing the main role of the premium-adjusted net delta, but it may understate the practical value of a richer pricing model in environments where curvature, tail events, or more complex rebalancing policies matter. A third limitation is that calibration was conducted snapshot by snapshot with smoothing, rather than in a fully dynamic state-space framework that incorporates the time-series structure of the snapshot-by-snapshot panel dataset.

These limitations point naturally to several extensions. A first extension is to evaluate richer hedging strategies, especially delta–gamma hedging. Such strategies would make use of more of the cross-sectional information contained in the option surface and may provide a setting in which the advantages of a richer model such as SVCJ become more visible in risk-management terms. A second extension is to move beyond the current affine specifications and consider more complex models that may be especially relevant in crypto markets, including (lifted) rough volatility and models with stochastic jump intensity. Rough volatility structures can better capture persistence and short-end behaviour in implied volatility, while stochastic jump intensity can account for clustering in discontinuous moves and changing tail-risk regimes. A third extension is to replace repeated snapshot-by-snapshot calibration with filtering-based estimation that calibrates option parameters to panel data on option prices in a formally dynamic way. Such an approach could improve parameter identification, reduce spurious instability, and provide a cleaner link between latent state dynamics and observed option surfaces.

In sum, the thesis delivers three main conclusions. First, a crypto-specific affine stochastic-volatility model with jumps materially improves the pricing of Deribit inverse options relative to Black and Heston, especially in the difficult parts of the surface. Second, inverse options should be analysed in a framework that takes their denomination and hedge mechanics seriously; the premium-adjusted net delta is not a minor correction, but a central object for hedging, giving consistently superior hedging results over the regular model delta. Third, better pricing does not automatically imply materially better first-order delta-hedging once one accounts for the basis risk inherent in using perpetual futures as the hedge instrument. These conclusions help clarify the respective roles of model richness and contract-specific hedge design in inverse crypto derivatives markets.

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A. Detailed Explanation of the Carr–Madan FFT Procedure

This appendix provides a detailed explanation of the Carr–Madan Fast Fourier Transform (FFT) method for pricing European options, following Carr and Madan (1999). The method computes option prices over a grid of strikes efficiently when the characteristic function of the log-underlying is available in (semi-)closed form, as in affine (jump-)diffusion models.

A.1 Foundations

Let F_T be the underlying futures price at maturity T, and let $X_T := \ln F_T$ denote the log-price. Under a pricing measure $\mathbb{Q}$ with deterministic discounting (constant short rate r for simplicity), the time-0 price of a European call with strike K is

$$C(K,T) = \mathbb{E}^{\mathbb{Q}}\big[e^{-rT}(F_T - K)^+\big] = e^{-rT}\int_{-\infty}^{\infty} (e^x - K)^+ f_{X_T}(x)\,\mathrm{d}x,$$

where f_{X_T} is the $\mathbb{Q}$-density of X_T. (In settings where discounting is negligible, one may set $r = 0$.)

Treat the call price as a function of log-strike $k := \ln K$. Then $\lim_{k\to-\infty} C(e^k, T) = e^{-rT}\mathbb{E}^{\mathbb{Q}}[F_T] = F_0$ (in the present futures setting), so $C(e^k, T)$ is not square-integrable in k over $\mathbb{R}$ and cannot be Fourier-transformed directly.

Following Carr and Madan (1999), introduce an exponential damping factor $\alpha > 0$ and define the *damped call price*

$$c(k,T) := e^{\alpha k}\, C(e^k, T). \tag{A.1}$$

For suitable α, $c(\cdot, T)$ is square-integrable. A sufficient condition is existence of the $(\alpha+1)$-moment:

$$\mathbb{E}^{\mathbb{Q}}[F_T^{\alpha+1}] < \infty,$$

which ensures $c(k,T) \to 0$ as $k \to \pm\infty$ and justifies Fourier inversion.

A.2 Fourier transform of the damped price

Define the Fourier transform of $c(\cdot, T)$ by

$$\widehat{c}(v,T) := \int_{-\infty}^{\infty} e^{ivk}\, c(k,T)\,\mathrm{d}k, \qquad v \in \mathbb{R}.$$

Using (A.1) and writing the call payoff integral in terms of k gives

$$\widehat{c}(v,T) = \int_{-\infty}^{\infty} e^{ivk} e^{\alpha k} \left(e^{-rT} \int_{k}^{\infty} (e^x - e^k)\, f_{X_T}(x)\,\mathrm{d}x \right) \mathrm{d}k.$$

Under the moment condition above, Fubini's theorem permits exchanging the order of integration:

$$\widehat{c}(v,T) = e^{-rT} \int_{-\infty}^{\infty} f_{X_T}(x) \left(\int_{-\infty}^{x} e^{(\alpha+iv)k} \left(e^x - e^k\right) \mathrm{d}k \right) \mathrm{d}x.$$

Evaluate the inner integral:

$$\begin{aligned}
\int_{-\infty}^{x} e^{(\alpha+iv)k}(e^x - e^k)\,\mathrm{d}k &= e^x \int_{-\infty}^{x} e^{(\alpha+iv)k}\,\mathrm{d}k - \int_{-\infty}^{x} e^{(\alpha+1+iv)k}\,\mathrm{d}k \\
&= e^x \frac{e^{(\alpha+iv)x}}{\alpha+iv} - \frac{e^{(\alpha+1+iv)x}}{\alpha+1+iv} \\
&= \frac{e^{(\alpha+1+iv)x}}{(\alpha+iv)(\alpha+1+iv)}.
\end{aligned}$$

Therefore,

$$\widehat{c}(v,T) = \frac{e^{-rT}}{(\alpha+iv)(\alpha+1+iv)} \int_{-\infty}^{\infty} e^{(\alpha+1+iv)x} f_{X_T}(x)\,\mathrm{d}x = \frac{e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\left[e^{(\alpha+1+iv)X_T}\right]}{(\alpha+iv)(\alpha+1+iv)}.$$

Let $\phi_T(u) := \mathbb{E}^{\mathbb{Q}}[e^{iuX_T}]$ denote the characteristic function of X_T. Since $e^{(\alpha+1+iv)X_T} = e^{i(v-i(\alpha+1))X_T}$, we obtain

$$\widehat{c}(v,T) = \frac{e^{-rT}\,\phi_T(v - i(\alpha+1))}{(\alpha+iv)(\alpha+1+iv)} = \frac{e^{-rT}\,\phi_T(v - i(\alpha+1))}{\alpha^2+\alpha-v^2+i(2\alpha+1)v}, \tag{A.2}$$

where we used $(\alpha+iv)(\alpha+1+iv) = \alpha^2+\alpha-v^2+i(2\alpha+1)v$.

A.3 Inversion and the Carr–Madan pricing formula

By Fourier inversion,

$$c(k,T) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ivk}\,\widehat{c}(v,T)\,\mathrm{d}v.$$

Since $c(k,T)$ is real-valued, the integral can be written in the one-sided form

$$c(k,T) = \frac{1}{\pi} \int_{0}^{\infty} \Re\!\left(e^{-ivk}\,\widehat{c}(v,T)\right) \mathrm{d}v.$$

Finally, undo the damping $c(k,T) = e^{\alpha k} C(e^k,T)$ to obtain

$$C(K,T) = \frac{e^{-\alpha k}}{\pi} \int_{0}^{\infty} \Re\!\left(e^{-ivk}\,\widehat{c}(v,T)\right) \mathrm{d}v, \qquad k = \ln K, \tag{A.3}$$

with $\widehat{c}(v,T)$ given explicitly by (A.2). Equations (A.2)–(A.3) are the Carr–Madan Fourier pricing representation (Carr and Madan, 1999).

A.4 Discretization and the FFT

To compute (A.3) numerically, truncate the integral and discretize the Fourier variable. Choose N (typically a power of 2 for FFT efficiency) and define a grid

$$v_j := j\eta, \qquad j = 0, 1, \dots, N-1,$$

with spacing $\eta > 0$. Approximating the integral by a weighted Riemann/trapezoidal sum yields

$$c(k, T) \approx \frac{1}{\pi} \sum_{j=0}^{N-1} w_j \, \Re\big(e^{-iv_j k} \, \widehat{c}(v_j, T)\big) \, \eta,$$

where w_j are numerical quadrature weights (e.g. trapezoidal weights $w_0 = w_{N-1} = 1/2$ and $w_j = 1$ otherwise; Alternatively, Simpson's rule may be used, with weights $w_0 = w_{N-1} = 1$, $w_j = 4$ for odd j, and $w_j = 2$ for even $j \in \{2, \dots, N-2\}$, which yields higher-order accuracy for smooth integrands, reducing the discretization error.).

To exploit the FFT, compute prices on an evenly spaced log-strike grid

$$k_u := b + u\lambda, \qquad u = 0, 1, \dots, N-1,$$

where $\lambda > 0$ is the log-strike spacing and b sets the minimum log-strike. Impose the FFT compatibility condition which synchronizes the strike grid and frequency grid so that the inverse Fourier transform reduces to a discrete Fourier transform:

$$\lambda\eta = \frac{2\pi}{N}.$$

Then

$$c(k_u, T) \approx \frac{1}{\pi} \Re\left(\sum_{j=0}^{N-1} e^{-iv_j(b+u\lambda)} \, \widehat{c}(v_j, T) \, w_j \, \eta \right) = \frac{1}{\pi} \Re\left(\sum_{j=0}^{N-1} e^{-iv_j b} \, e^{-i\frac{2\pi}{N} ju} \, \widehat{c}(v_j, T) \, w_j \, \eta \right).$$

Define the sequence

$$x_j := e^{-iv_j b} \, \widehat{c}(v_j, T) \, w_j \, \eta, \qquad j = 0, \dots, N-1.$$

Then the bracketed sum is precisely the discrete Fourier transform (DFT) of (x_j) evaluated at index u, which can be computed for all u in $O(N \log N)$ time via the FFT:

$$c(k_u, T) \approx \frac{1}{\pi} \Re\big(\mathrm{DFT}(x)_u\big), \qquad C(e^{k_u}, T) = e^{-\alpha k_u} c(k_u, T).$$

In our application, $\phi_T(\cdot)$ is provided by the model characteristic function derived in the main text (e.g. Section 2.2.3), evaluated at the complex argument $v - i(\alpha + 1)$ as required by (A.2).

A.5 Practical implementation considerations

- **Choice of damping factor α.** The parameter $\alpha > 0$ must be chosen so that $\mathbb{E}^{\mathbb{Q}}[F_T^{\alpha+1}] < \infty$, equivalently, so that $\phi_T(v - i(\alpha + 1))$ exists for the required range of v. In models with jumps, this can impose nontrivial restrictions on α (through exponential moments of the jump component). Practical guidance on admissible damping and numerical stability is discussed in Carr and Madan (1999) and Lee (2004).

- **Truncation/discretization parameters.** The truncation level $v_{\max} = (N - 1)\eta$ should be large enough to control truncation error, and η should be small enough to control discretization error. In practice one selects (N, η) to cover the strike range of interest via $\lambda = 2\pi/(N\eta)$ and checks numerical stability by refinement.

- **Characteristic function evaluation.** Care is required when evaluating $\phi_T(u)$ at complex arguments, in particular in models where ϕ_T involves complex square roots and logarithms (e.g. Heston/SVCJ Riccati solutions). A consistent branch choice (typically enforcing $\Re(d(u)) > 0$ for the Heston discriminant) helps avoid numerical discontinuities.

This FFT-based approach yields an efficient way to compute a full grid of call prices across strikes, which is particularly useful for calibration to cross-sectional option data.

B. Characteristic functions under the futures price numeraire for the Black and Heston models

This appendix derives the conditional characteristic functions of $X_T = \log F_T$ given (X_t, V_t) (or X_t alone in the Black model) under the *futures price numeraire.* In both cases we work under the pricing measure for which the futures price $(F_t)_{t\geq 0}$ is a martingale, i.e.

$$\mathbb{E}\left[\frac{\mathrm{d}F_t}{F_t}\Big|\mathcal{F}_t\right] = 0.$$

Equivalently, for $X_t = \log F_t$ the drift is chosen so that $F_t = e^{X_t}$ has zero drift in expectation.

B.1 Black model

Model. Under the futures price numeraire, the Black model assumes

$$\mathrm{d}F_t = \sigma F_t \,\mathrm{d}W_t, \qquad \sigma > 0, \tag{B.1}$$

so that F_t is a martingale. Let $X_t := \log F_t$. By Itô's formula,

$$\mathrm{d}X_t = -\frac{1}{2}\sigma^2\,\mathrm{d}t + \sigma\,\mathrm{d}W_t. \tag{B.2}$$

Conditional characteristic function. Fix $u \in \mathbb{C}$ and define

$$\phi_{\mathrm{BL}}(u;t,T) := \mathbb{E}\big[e^{iuX_T} \mid \mathcal{F}_t\big]\,.$$

From (B.2),

$$X_T = X_t - \frac{1}{2}\sigma^2(T-t) + \sigma\,(W_T - W_t),$$

and $W_T - W_t \sim \mathcal{N}(0, T-t)$ is independent of $\mathcal{F}_t$. Hence

$$\begin{aligned}\phi_{\mathrm{BL}}(u;t,T) &= \exp\Big(iu\big(X_t - \tfrac{1}{2}\sigma^2(T-t)\big)\Big)\,\mathbb{E}\big[e^{iu\sigma(W_T-W_t)}\big]\\ &= \exp\Big(iuX_t - \tfrac{1}{2}iu\sigma^2\tau\Big)\,\exp\Big(-\tfrac{1}{2}u^2\sigma^2\tau\Big), \qquad \tau := T-t.\end{aligned} \tag{B.3}$$

Therefore,

$$\boxed{\phi_{\mathrm{BL}}(u;t,T) = \exp\bigg(iuX_t - \frac{1}{2}\sigma^2\tau\,(u^2+iu)\bigg)}\,. \tag{B.4}$$

B.2 Heston model

Model. Under the futures price numeraire, the Heston model assumes

$$\mathrm{d}F_t = F_t\sqrt{V_t}\,\mathrm{d}W_{1t}, \tag{B.5}$$

$$\mathrm{d}V_t = \kappa(\theta - V_t)\,\mathrm{d}t + \sigma_v\sqrt{V_t}\,\mathrm{d}W_{2t}, \tag{B.6}$$

with $\kappa > 0$, $\theta > 0$, $\sigma_v > 0$, and $\mathrm{corr}(\mathrm{d}W_{1t}, \mathrm{d}W_{2t}) = \rho$. Let $X_t := \log F_t$. By Itô's formula applied to $F_t = e^{X_t}$,

$$\mathrm{d}X_t = -\frac{1}{2}V_t\,\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_{1t}. \tag{B.7}$$

The drift $-\frac{1}{2}V_t$ is the unique choice ensuring that F_t is a martingale under this measure.

Martingale ansatz. Fix $u \in \mathbb{C}$ and define the conditional characteristic function

$$\phi_{\mathrm{H}}(u; t, T) := \mathbb{E}\left[e^{iuX_T} \mid X_t, V_t\right].$$

Motivated by the affine structure of (B.7)–(B.6), we conjecture the exponential-affine form

$$\phi_{\mathrm{H}}(u; t, T) = \exp\big(A_{\mathrm{H}}(\tau; u) + B_{\mathrm{H}}(\tau; u)V_t + iuX_t\big), \qquad \tau := T - t, \tag{B.8}$$

with terminal conditions $A_{\mathrm{H}}(0; u) = 0$ and $B_{\mathrm{H}}(0; u) = 0$.

Define $\Phi_t := \exp(A_{\mathrm{H}}(\tau; u) + B_{\mathrm{H}}(\tau; u)V_t + iuX_t)$. As in the SVCJ derivation in Section 2.2, we choose A_{H} and B_{H} so that $(\Phi_t)_{t\in[0,T]}$ is a martingale. Applying Itô's formula to $\Phi(t, X_t, V_t)$ and collecting the predictable drift terms yields a drift that is affine in V_t. Setting this drift equal to zero for all (X_t, V_t) produces the Riccati system

$$\frac{\partial}{\partial\tau}B_{\mathrm{H}}(\tau; u) = \frac{1}{2}\sigma_v^2 B_{\mathrm{H}}(\tau; u)^2 + (\rho\sigma_v iu - \kappa)\,B_{\mathrm{H}}(\tau; u) - \frac{1}{2}(u^2 + iu), \qquad B_{\mathrm{H}}(0; u) = 0, \tag{B.9}$$

$$\frac{\partial}{\partial\tau}A_{\mathrm{H}}(\tau; u) = \kappa\theta\,B_{\mathrm{H}}(\tau; u), \qquad A_{\mathrm{H}}(0; u) = 0. \tag{B.10}$$

Closed-form solution. Define

$$d(u) := \sqrt{\big(\kappa - \rho\sigma_v iu\big)^2 + \sigma_v^2\big(u^2 + iu\big)}, \tag{B.11}$$

$$g(u) := \frac{\kappa - \rho\sigma_v iu - d(u)}{\kappa - \rho\sigma_v iu + d(u)}, \qquad \Re(d(u)) > 0. \tag{B.12}$$

Then (B.9) admits the closed-form solution (Heston, 1993)

$$\boxed{B_{\mathrm{H}}(\tau; u) = \frac{\kappa - \rho\sigma_v iu - d(u)}{\sigma_v^2}\,\frac{1 - e^{-d(u)\tau}}{1 - g(u)e^{-d(u)\tau}},} \tag{B.13}$$

and integrating (B.10) gives (Heston, 1993)

$$\boxed{A_{\mathrm{H}}(\tau; u) = \frac{\kappa\theta}{\sigma_v^2}\left[\big(\kappa - \rho\sigma_v iu - d(u)\big)\tau - 2\log\left(\frac{1 - g(u)e^{-d(u)\tau}}{1 - g(u)}\right)\right].} \tag{B.14}$$

Result. Therefore, the Heston characteristic function under the futures price numeraire is

$$\boxed{\phi_{\mathrm{H}}(u;t,T) = \exp\big(A_{\mathrm{H}}(\tau;u) + B_{\mathrm{H}}(\tau;u)V_t + iuX_t\big), \qquad \tau = T - t,} \tag{B.15}$$

with A_{H} and B_{H} given by (B.14) and (B.13).

C. Proof of Put–Call Parity for Inverse Options under the Futures Price Numeraire

This appendix proves the put–call parity stated in Theorem 2.3.11 in the main text. Throughout, we assume zero discounting (equivalently, a zero interest rate) and a frictionless market.

C.1 Payoff identity

Let F_T denote the futures price at maturity T and fix a strike $K > 0$. The European call and put payoffs at T are

$$(F_T - K)^+, \qquad (K - F_T)^+.$$

For every real number x, the identity $(x)^+ - (-x)^+ = x$ holds. Applying it with $x = F_T - K$ yields the pointwise payoff equality

$$(F_T - K)^+ - (K - F_T)^+ = F_T - K. \tag{C.1}$$

Therefore, a portfolio that is long one call and short one put (same K, T) replicates the forward-type payoff $F_T - K$.

C.2 Pricing the forward-type payoff with a futures contract (zero discounting)

Let $C_t(K, T)$ and $P_t(K, T)$ be the time-t prices (in USD units) of the call and put. By linearity of pricing,

$$C_t(K,T) - P_t(K,T) \quad \text{is the time-}t\text{ value of a claim paying} \quad F_T - K \text{ at } T. \tag{C.2}$$

Under zero discounting, the payoff $F_T - K$ can be replicated using (i) a long position in one futures contract and (ii) a static cash position:

- Enter a long futures position at time t. With standard mark-to-market accounting and zero interest, the cumulative gain from t to T is $F_T - F_t$.

- Hold a cash amount of $(F_t - K)$ from t to T (no growth under zero discounting).

The terminal value of this self-financing strategy is

$$(F_t - K) + (F_T - F_t) = F_T - K,$$

which matches the desired payoff. Hence the no-arbitrage time-t value of the claim $F_T - K$ equals the initial cost of this replicating strategy, namely $F_t - K$:

$$C_t(K,T) - P_t(K,T) = F_t - K. \tag{C.3}$$

C.3 Parity in futures-numeraire (coin) units

Define the futures-numeraire (coin) prices

$$\widehat{C}_t(K,T) := \frac{C_t(K,T)}{F_t}, \qquad \widehat{P}_t(K,T) := \frac{P_t(K,T)}{F_t}.$$

Dividing (C.3) by F_t gives

$$\boxed{\widehat{C}_t(K,T) - \widehat{P}_t(K,T) = \frac{F_t - K}{F_t} = 1 - \frac{K}{F_t},} \tag{C.4}$$

which is exactly the parity stated in (2.3.11).

D. Regular Deltas under the Black, Heston and SVCJ Models for Inverse Options

This appendix derives the *regular* deltas used in the empirical hedging analysis. The implementation first computes the delta of the corresponding regular USD-settled option with respect to the current dated futures price, and only afterwards converts it into the premium-adjusted net delta used for inverse-option hedging.

Let F_t denote the current dated futures price, let K denote the strike, let $\tau := T - t$ denote time to maturity, and let

$$X_T := \log F_T.$$

Write the regular USD call and put prices as $C_t = C(F_t, K, \tau)$ and $P_t = P(F_t, K, \tau)$. The regular deltas are

$$\Delta^{\text{reg}}_{\text{call}} := \frac{\partial C(F_t, K, \tau)}{\partial F_t}, \qquad \Delta^{\text{reg}}_{\text{put}} := \frac{\partial P(F_t, K, \tau)}{\partial F_t}.$$

The inverse option price in coin units is

$$V^{\text{coin}}_t = \frac{V^{\text{USD}}_t}{F_t},$$

and the premium-adjusted net delta used in the main text is

$$\Delta^{\text{net}}_t = \Delta^{\text{reg}}_t - V^{\text{coin}}_t.$$

This appendix focuses only on the derivation of Δ^{reg}.

D.1 General representation

Work under the futures-price measure $\mathbb{Q}^F$, under which F_t is a martingale. Let

$$\phi(u) := \mathbb{E}^{\mathbb{Q}^F}\left[e^{iuX_T} \mid \mathcal{F}_t\right]$$

denote the conditional characteristic function of X_T. Since $F_T = e^{X_T}$,

$$\phi(-i) = \mathbb{E}^{\mathbb{Q}^F}\left[e^{X_T} \mid \mathcal{F}_t\right] = \mathbb{E}^{\mathbb{Q}^F}\left[F_T \mid \mathcal{F}_t\right] = F_t.$$

For a regular call option,

$$C_t = F_t\Pi_{1,t} - K\Pi_{2,t},$$

where

$$\Pi_{1,t} = \mathbb{Q}^F(F_T > K \mid \mathcal{F}_t) = \mathbb{Q}^F(X_T > \log K \mid \mathcal{F}_t),$$

and

$$\Pi_{2,t} = F_t\, \mathbb{E}^{\mathbb{Q}^F}\left[\frac{\mathbf{1}_{\{F_T>K\}}}{F_T} \,\middle|\, \mathcal{F}_t \right].$$

For convenience, define the auxiliary measure $\widetilde{\mathbb{Q}}$ by

$$\left.\frac{d\widetilde{\mathbb{Q}}}{d\mathbb{Q}^F}\right|_{\mathcal{F}_T} = \frac{F_T}{F_t} = \frac{e^{X_T}}{F_t}.$$

Then

$$\Pi_{2,t} = \widetilde{\mathbb{Q}}(F_T > K \mid \mathcal{F}_t) = \widetilde{\mathbb{Q}}(X_T > \log K \mid \mathcal{F}_t).$$

Under zero discounting, $\widetilde{\mathbb{Q}}$ coincides with the usual cash-numeraire pricing measure, but that interpretation is not needed below.

By the Gil–Pelaez inversion formula (Gil-Pelaez, 1951),

$$\Pi_{1,t} = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \Re\left(\frac{e^{-iu\log K}\phi(u)}{iu} \right) du, \tag{C.1}$$

$$\begin{aligned} \Pi_{2,t} &= \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \Re\left(\frac{e^{-iu\log K}\phi(u-i)}{iu\,\phi(-i)} \right) du \\ &= \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \Re\left(\frac{e^{-iu\log K}\phi(u-i)}{iu\,F_t} \right) du. \end{aligned} \tag{C.2}$$

Indeed, the characteristic function of X_T under $\widetilde{\mathbb{Q}}$ is

$$\widetilde{\phi}(u) = \mathbb{E}^{\widetilde{\mathbb{Q}}}\left[e^{iuX_T} \mid \mathcal{F}_t \right] = \frac{\phi(u-i)}{\phi(-i)} = \frac{\phi(u-i)}{F_t}.$$

Under standard regularity conditions, the regular call delta is

$$\Delta_{\text{call}}^{\text{reg}} = \Pi_{1,t},$$

and the regular put delta follows from put–call parity,

$$P_t = C_t - F_t + K,$$

hence

$$\Delta_{\text{put}}^{\text{reg}} = \Delta_{\text{call}}^{\text{reg}} - 1 = \Pi_{1,t} - 1.$$

Thus the empirical hedging engine computes call deltas first, and obtains put deltas by subtracting one.

D.2 Black model

Under the Black model, under the futures-price measure,

$$dF_t = \sigma F_t \, dW_t,$$

so

$$dX_t = -\frac{1}{2}\sigma^2 \, dt + \sigma \, dW_t.$$

Therefore

$$X_T \sim \mathcal{N}\left(\log F_t - \frac{1}{2}\sigma^2\tau,\ \sigma^2\tau\right).$$

The regular call price is

$$C_t = F_t N(d_1) - K N(d_2),$$

where

$$d_1 = \frac{\log(F_t/K) + \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}.$$

Hence

$$\Delta^{\text{reg}}_{\text{call,Black}} = N(d_1), \tag{C.3}$$

$$\Delta^{\text{reg}}_{\text{put,Black}} = N(d_1) - 1. \tag{C.4}$$

Equivalently,

$$\Pi^{\text{Black}}_{1,t} = N(d_1), \qquad \Pi^{\text{Black}}_{2,t} = N(d_2).$$

D.3 Heston model

Under the Heston model, under the futures-price measure,

$$dF_t = F_t \sqrt{V_t} \, dW_{1,t}, \tag{C.5}$$

$$dV_t = \kappa(\theta - V_t) \, dt + \sigma_v \sqrt{V_t} \, dW_{2,t}, \tag{C.6}$$

with

$$\operatorname{corr}(dW_{1,t}, dW_{2,t}) = \rho.$$

Equivalently, for $X_t = \log F_t$,

$$dX_t = -\frac{1}{2} V_t \, dt + \sqrt{V_t} \, dW_{1,t}.$$

The conditional characteristic function is exponential-affine:

$$\phi_H(u) = \mathbb{E}^{\mathbb{Q}^F}\left[e^{iuX_T} \mid X_t, V_t\right] = \exp\Big(A_H(\tau; u) + B_H(\tau; u) V_t + iuX_t\Big),$$

where (Heston, 1993)

$$B_H(\tau;u) = \frac{\kappa - \rho\sigma_v iu - d(u)}{\sigma_v^2} \cdot \frac{1 - e^{-d(u)\tau}}{1 - g(u)e^{-d(u)\tau}}, \tag{C.7}$$

$$A_H(\tau;u) = \frac{\kappa\theta}{\sigma_v^2}\left[\big(\kappa - \rho\sigma_v iu - d(u)\big)\tau - 2\log\!\left(\frac{1 - g(u)e^{-d(u)\tau}}{1 - g(u)}\right)\right], \tag{C.8}$$

with

$$d(u) = \sqrt{(\kappa - \rho\sigma_v iu)^2 + \sigma_v^2(u^2 + iu)}, \qquad g(u) = \frac{\kappa - \rho\sigma_v iu - d(u)}{\kappa - \rho\sigma_v iu + d(u)}.$$

Using (C.1), the regular call delta is

$$\Delta^{\text{reg}}_{\text{call,Heston}} = \Pi^H_{1,t} = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \Re\!\left(\frac{e^{-iu\log K}\phi_H(u)}{iu}\right) du,$$

and the regular put delta is

$$\Delta^{\text{reg}}_{\text{put,Heston}} = \Delta^{\text{reg}}_{\text{call,Heston}} - 1.$$

So in the Heston case the implementation computes the regular delta directly from the $\Pi_{1,t}$ Fourier integral, rather than by finite-differencing the FFT price grid.

D.4 SVCJ model

Under the stochastic-volatility-with-correlated-jumps model,

$$dX_t = \left(-\frac{1}{2}V_t - \lambda\kappa_F\right) dt + \sqrt{V_t}\, dW_{1,t} + Z^y\, dN_t, \tag{C.9}$$

$$dV_t = \kappa(\theta - V_t)\, dt + \sigma_v\sqrt{V_t}\, dW_{2,t} + Z^v\, dN_t, \tag{C.10}$$

with

$$\operatorname{corr}(dW_{1,t}, dW_{2,t}) = \rho,$$

and jump structure

$$Z^v \sim \operatorname{Exp}(\text{mean } \ell_v), \qquad Z^y \mid Z^v \sim \mathcal{N}\big(\ell_y + \rho_j Z^v, \sigma_y^2\big).$$

The drift adjustment is

$$\kappa_F = \mathbb{E}\big[e^{Z^y} - 1\big] = \exp\!\left(\ell_y + \frac{1}{2}\sigma_y^2\right)\frac{1}{1 - \ell_v\rho_j} - 1.$$

The conditional characteristic function is again exponential-affine:

$$\phi_J(u) = \mathbb{E}^{\mathbb{Q}^F}\left[e^{iuX_T} \mid X_t, V_t\right] = \exp\!\Big(A_J(\tau;u) + B_J(\tau;u)V_t + iuX_t\Big),$$

with

$$B_J(\tau; u) = B_H(\tau; u),$$

and

$$A_J(\tau; u) = A_J^{\text{diff}}(\tau; u) + A_J^{\text{jump}}(\tau; u),$$

where

$$A_J^{\text{diff}}(\tau; u) = \frac{\kappa\theta}{\sigma_v^2}\left[\big(\kappa - \rho\sigma_v iu - d(u)\big)\tau - 2\log\left(\frac{1 - g(u)e^{-d(u)\tau}}{1 - g(u)}\right)\right],$$

and

$$A_J^{\text{jump}}(\tau; u) = \lambda \int_0^\tau \Big(M\big(u, B_J(s; u)\big) - 1 - iu\kappa_F\Big)\, ds,$$

with

$$M(u, B) = \mathbb{E}\big[e^{iuZ^y + BZ^v}\big] = \exp\left(iu\ell_y - \frac{1}{2}u^2\sigma_y^2\right)\frac{1}{1 - \ell_v(B + iu\rho_j)}.$$

See Chapter 2 for the full characteristic-function derivation.

Using (C.1), the regular call delta is

$$\Delta_{\text{call,SVCJ}}^{\text{reg}} = \Pi_{1,t}^J = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \Re\left(\frac{e^{-iu\log K}\phi_J(u)}{iu}\right) du,$$

and the regular put delta is

$$\Delta_{\text{put,SVCJ}}^{\text{reg}} = \Delta_{\text{call,SVCJ}}^{\text{reg}} - 1.$$

Thus, just as in the Heston case, the SVCJ regular delta is computed as a semi-analytic Fourier quantity from the characteristic function, rather than as a numerical derivative of the FFT price surface.